An Accounting-based Asset Pricing Model and a Fundamental Factor

Stephen Penman

Columbia Business School, Columbia University and Bocconi University

Julie Zhu

Fanhai International School of Finance, Fudan University

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Stephen Penman is at shp38@columbia.edu and Julie Zhu is at leizhu@fudan.edu.cn

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Abstract

This paper recasts the consumption asset pricing model in terms of accounting numbers that connect to consumption and the risk to consumption under accounting principles. The modeling yields an expected return measure that forecasts realized returns and the risk to those returns. It leads to the construction of a pricing factor from the accounting information. The factor performs well relative to extant factors in explaining cross-sectional returns. The factor return has negative correlation with the market portfolio and exhibits the property of protecting payoffs in bad states when consumption is low. This then prompts a two-factor representation that combines the market portfolio with a hedge portfolio against loss to consumption.

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1. Introduction

This paper develops an asset pricing model that recasts the consumption-based pricing model in terms of observable accounting outcomes. The model leads to the construction of a fundamental factor model which is then compared to extant factor models. Many of those models incorporate accounting numbers such as book-to-price, various profitability measures, and investment, but these have largely been obtained in data driven exercises. In contrast, the accounting numbers underlying the fundamental factor are identified from consumption-based asset pricing theory overlaid with accounting principles that link those numbers to consumption and the risk to consumption.

Two accounting principles make the links. First, the clean-surplus accounting relation: Dividends that buy consumption are paid out of book value, and earnings add to book value. Thus, it is future book values and the earnings that generate them that are at risk. Second, the realization principle for booking earnings: Earnings are booked when risk is resolved. Consequently, (unrealized) expected earnings indicate future dividends and consumption at risk, while realized earnings inform that consumption risk is resolved. These are the accounting features that tie to risk in a general, no-arbitrage pricing model in Penman and Zhang (2020). The pricing model and the associated factor model here follow from that paper.

As with other factor models, the factor prompted by the model is a long-short, zero-net-investment portfolio, but one with a higher mean return and a higher Sharpe ratio. Further, the factor exhibits alphas relative to returns explained by other factor models, while the opposite is

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largely not the case. Indeed, the fundamental factor parsimoniously captures most of the returns explained by those other models in one factor rather than the "zoo of factors" that are added from data extraction.

Being based on theory and observables, the factor is interpretable and transparent, in contrast to factors such as "value" and "size" and even accounting-based factors such as the "profitability factor" and the "CMA factor" that lack clear attribution to risk and return. Thus, the fundamental factor serves as a benchmark for evaluating other factors on the extent to which they capture fundamental risk. That breaks the skepticism of Lewellen, Nagel, and Shanken (2010) about attempts to explain factors simply via correlations and R^2. More so, the benchmarking is with a single factor rather than many factors that are more likely to capture factor structure to increase R^2. Connected to the risk to consumption, the factor is also likely to capture risk that investors are concerned about and less likely to be correlated with the payoffs that should not be priced.

An objective of asset pricing is not only to explain the cross-section of realized returns in-sample but to yield an expected return in real time (out of sample) for investors in practice, the point emphasized by Black (1993).[1] Our analysis not only yields an out-of-sample expected return that forecasts actual returns but also the risk to returns investors actually experience, as well as the risk to earnings from which the dividends (that buy consumption) are paid.

Interestingly, the fundamental factor has negative beta with respect to the market portfolio. Further, it exhibits the property of protecting payoffs in bad states when markets are down and presumably consumption is low, a key feature in consumption-based asset pricing theory. These properties motivate a two-factor model with a market factor and a protective portfolio against

[1] "The key issue in investments is *estimating* the expected return. It is neither explaining return…nor explaining average return" (Black 1993, emphasis in the original).

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loss to consumption, a portfolio that hedges intertemporal uncertainty to consumption-investment opportunities conveyed by state variables, as in Merton (1973) and Campbell (1993).

Beside the empirical confirmation, the persuasion for the model comes from its grounding in consumption-based asset pricing theory and from an understanding of accounting principles that connect accounting numbers to consumption at risk. The paper begins by making the connections. It then estimates firm-specific expected returns implied by the relevant accounting and constructs the fundamental risk factor, reports its properties, and compares it to alternative factor models. Finally, the paper explores the properties of a two-factor model that incorporates the fundamental factor along with the market portfolio.

2. Connecting Accounting to Consumption and the Risk to Consumption

The logic runs as follows. First, as with standard consumption-based asset pricing theory, the risk to investing is the risk to future consumption that the investment is expected to yield. Second, dividends buy consumption and dividends are paid out of book value. So, it is the risk to future book value that is the risk to consumption. Third, book value is generated by earnings, so the risk surrounds future earnings that add to book value. Fourth, accounting principles that govern the booking of earnings deal with risk and its resolution, and tie the risk to earnings to the risk to consumption. Indeed, those accounting principles tie to the discount factor in Penman and Zhang (2020). Fifth, accounting numbers generated under those principles convey this priced risk, and it is those accounting numbers from which a risk factor is constructed.

2.1 Accounting that Links to Consumption

Under standard consumption-based asset pricing theory emanating from Rubinstein (1976), the pricing of future dividends, d_τ, that buy future consumption is given by

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$$P_t = E_t \sum_{\tau=t}^{\infty} m_\tau d_\tau$$

$$= \sum_{\tau=t+1}^{\infty} \frac{E_t(d_\tau)}{R_\tau^f} + \sum_{\tau=t+1}^{\infty} \text{cov}_t(d_\tau, m_\tau) \qquad (1)$$

provided $E_t(P_\tau m_\tau) \to 0$ as $\tau \to \infty$. (Variables subscripted $\tau > t$ are expected values conditional on information at time t.) $m_\tau = \beta \frac{u'(c_\tau)}{u'(c_t)} > 0$ which exists given no arbitrage, β is an impatience factor, R_τ^f is one plus the risk-free discount rate for period, τ, and $\frac{u'(c_\tau)}{u'(c_t)}$ the marginal rate of substitution of time-τ consumption for current consumption, determined by the relative marginal utility from consumption at those dates.[2] The investor invests to equate the utility lost from surrendering current consumption with that gained, and views the variation of future consumption (and the utility from consumption) as risk—thus the covariance term. With decreasing marginal utility with increasing consumption, payoffs which vary positively with marginal utility (and thus have higher payoffs when consumption is low) have higher prices (and lower expected returns). This feature is important for interpreting our empirical analysis.

As it stands, equation (1) is lacking content without m_τ identified; the separation of discounted dividends from the covariance term is simply a mathematical reformulation of $E_t(m_\tau d_\tau)$. More structure that conveys m_τ in terms of observables is required. Accounting supplies observable numbers and the paper asks how those numbers might provide the necessary identification.

[2] The question arises: Whose m_τ? The early theory relied on preferences aggregating to those of a representative average investor. Breeden and Litzenberger (1978) and Breeden (1979) provide theorems where, under certain assumptions, diverse preferences aggregate to a derive asset pricing models in terms of preference over aggregate consumption. Accounting is prescribed for all firms (the aggregate), with accounting principles derived from common practice over centuries, so can be viewed as a consensual accommodation of uncertainty.

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However, there is another issue. While price is given by discounted expected dividends with the infinite summation, P_t is insensitive to dividends up to any finite point, $\tau = T$, under Miller and Modigliani (1961) (M&M) conditions; the timing of dividend payments does not determine the timing of consumption as the investor can rearrange the cash flows to get the desired periodic consumption. Thus, the expectation and covariance term in equation (1) for $t < \tau \leq T$, all T, are of no interest to the investor. Valuation (and risk to consumption) must be based on something more "fundamental," capturing the generation of dividends rather than their distribution.

For a firm, that "something" comes from adding value by trading in input (supplier) and output (customer) markets. Accountants attempt to calculate this spread as $Earnings_\tau$ (revenues minus expenses). However, they do so in a double-entry system that also tracks shareholders' (book) equity, B_τ, the stock of equity as well as the flow, $Earnings_\tau$. The two articulate:

Accounting Principle 1: $B_\tau = B_{\tau\text{-}1} + Earnings_\tau - d_\tau$, all τ (CSR)

Under this "clean-surplus accounting relation," book value is generated by earnings, and dividends are paid out of the resulting book value. It is implicit that dividends do not affect earnings—they don't under U.S. GAAP and IFRS. So, importantly, the generation of value is distinguished from the distribution of value.[3] Dividends are net dividends—cash dividends plus share repurchases, minus share issues. So, the equation expresses the cash-to-cash cycle (consumption-to-consumption cycle): Cash paid into the firm (that surrenders current consumption) creates book value, earnings add to the book value, and cash (that is applied to future consumption) is paid out of this book value. Given current observed book value, B_t, CSR implies that expected future book value at any point T is

[3] $Earnings_t$ must be comprehensive earnings, and that is reported under U.S. GAAP and IFRS. Under those accounting regimes, comprehensive income is divided between net income and "other comprehensive income," but the latter is usually zero in expectation (for $\tau > t$).

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$$E_t(B_T) = B_t + \sum_{\tau=t+1}^{T} E_t(Earnings_\tau - d_\tau) \qquad \text{(CSRT)}$$

That is, with dividends paid out of book value, expected dividends that buy consumption at each point T are given by the expected book value at that point. That book value is generated by earnings up to that point, diminished by prior dividends already paid out. This is an accounting for expected dividends and thus expected consumption.

With dividends paid out of book value,

$$P_t^T = \sum_{\tau=t+1}^{T} \frac{E_t(d_\tau)}{R_\tau^f} + \frac{E_t(B_T)}{R_T^f} + \sum_{\tau=t+1}^{T} \text{cov}_t(d_\tau, m_\tau) + \text{cov}_t(B_T, m_T) \qquad (2)$$

$\rightarrow P_t$ in equation (1) as $T \rightarrow \infty$, provided $E_t(B_\tau m_\tau) \rightarrow 0$ as $T \rightarrow \infty$. Implicit in this condition, $\frac{E_t(B_T)}{R_T^f} \rightarrow 0$ as $T \rightarrow \infty$, that is, dividends are ultimately paid out of book value. This valuation establishes future book value (from which dividends are paid) as being at risk.[4]

By CSRT, expected book value, B_T, is generated by expected earnings (net of dividends) for each period, τ, up to T. So, setting $cov_t(B_T, m_T) = \sum_{\tau=t+1}^{T} cov_t[(Earnings_\tau - d_\tau), m_\tau]$, equation (2) can be restated as

$$P_t^T = \frac{B_t}{R_T^f} + \sum_{\tau=t+1}^{T} \frac{E_t(d_\tau)}{R_\tau^f} + \frac{\sum_{\tau=t+1}^{T} E_t(Earnings_\tau - d_\tau)}{R_T^f} + \sum_{\tau=t+1}^{T} \text{cov}_t(d_\tau, m_\tau) + \sum_{\tau=t+1}^{T} \text{cov}_t\left[(Earnings_\tau - d_\tau), m_\tau\right]$$

$$= \frac{B_t}{R_T^f} + \sum_{\tau=t+1}^{T} \frac{E_t(d_\tau)}{R_\tau^f} + \frac{\sum_{\tau=t+1}^{T} E_t(Earnings_\tau - d_\tau)}{R_T^f} + \sum_{\tau=t+1}^{T} \text{cov}_t(Earnings_\tau, m_\tau)$$

[4] Equation (2) can be expressed as the familiar residual income model. Feltham and Ohlson (1999) formulate that model with the discount for risk as the covariance of residual income with a pricing kernal.

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$$= \frac{B_t}{R_T^f} + \frac{1}{R_T^f} \sum_{\tau=t+1}^{T} E_t \left[Earnings_\tau + d_\tau \left(\frac{R_T^f}{R_\tau^f} - 1 \right) \right] + \sum_{\tau=t+1}^{T} \text{cov}_t (Earnings_\tau, m_\tau) \quad (3)$$

The covariance term expresses risk in terms of earnings at risk, that is, the extent to which earnings covary with the stochastic discount factor. Dividends effectively drop out in the last expression, as they should under M&M conditions. In the term, $d_\tau \left(\frac{R_T^f}{R_\tau^f} - 1 \right)$, expected dividends for each τ up to *T* are reinvested, representing the investor who wishes to delay consumption, and that reinvestment yields more earnings to add to the book value from which future dividends can be paid for the desired later consumption. With investment at the risk-free rate, consumption is preserved. If a dividend is not paid, but rather invested at the same risk-free rate (in "cash") within the firm, consumption is similarly preserved. For the investor who consumes the dividend, the dividend displaces future earnings and book value (and correspondingly reduces future consumption).[5] In short, the payment of dividends does not affect cum-dividend book value from which dividends are paid; the price in equation (3) that values consumption is "dividend payout irrelevant." Only investment that produces value-added earnings potentially affects price, and it is the covariance of those earnings with the stochastic discount factor in equation (3) that captures the risk to consumption, not the covariance with dividends in equation (1). But note that it is the accounting for earnings and book value that yields the dividend-irrelevant pricing. Just as dividends displace price dollar-for-dollar under M&M assumptions, so they displace book value dollar-for-dollar and do not affect current earnings (in CSR). Ohlson (1995) stresses the point in connecting accounting numbers to no-arbitrage prices.[6]

[5] Penman and Sougiannis (1997) document that this earnings displacement property is evident with GAAP.

[6] To see this more clearly, note that the finite-horizon valuation,

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2.2 Accounting that Links to the Risk to Consumption

However, something else must be put on the table. If accounting numbers are to indicate risk, they require further specification such that the covariance term involving earnings in equation (3), $cov_t(Earnings_\tau, m_\tau)$, actually captures the risk to consumption. That is satisfied by accounting principles that govern the measurement of earnings. By itself, CSR and equation (3) do not prescribe the relevant accounting. Indeed, random numbers can satisfy clean-surplus accounting, constrained only by the "transversality condition" in the limit operator; up to any point, T, earnings can be any arbitrary number unrelated to the risk to consumption. The covariance of m_τ with a random number has even less meaning than the covariance with dividends.

In short, periodic earnings added to book value in equation (3) must represent consumption value from which dividends can be paid and, correspondingly, the covariance term, $cov_t(Earnings_\tau, m_\tau)$ represents expected consumption at risk. What would be the determining accounting principles, and are they applied under U.S. GAAP or IFRS?

Under GAAP and IFRS,

Accounting Principle 2: Under uncertainty, the recognition of earnings is deferred to the future until the uncertainty is resolved.

$$P_t^T = \sum_{\tau=t+1}^{T} \frac{E_t(d_\tau)}{R_\tau^f} + \frac{E_t(P_T)}{R_T^f} + \sum_{\tau=t+1}^{T} \mathrm{cov}_t(d_\tau, m_\tau) + \mathrm{cov}_t(P_T, m_T)$$

is independent of d_τ: Expected dividends for $t < \tau \leq T$ displace $E_t(P_T)$ to leave P_t^T unchanged. Similarly, expected d_τ for $t < \tau \leq T$ displace $E_t(B_T)$ in equation (2) to leave P_t^T unchanged. Violation of M&M assumptions might modify the dividend irrelevance property, requiring a valuation modified with frictions. The point here is to emphasize that accounting works in accordance with a foundational property of asset pricing.

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This "realization principle," familiar from Accounting 101 class, not only prescribes when earnings are booked but links that determination to uncertainty: To book earnings, there must be little uncertainty that the earnings will generate cash to pay dividends. Accordingly, realized earnings are those that realize consumption. If not, earnings recognition is deferred to future periods until that uncertainty is resolved. Expected earnings added to book value in equation (3) are those that buy consumption but, as they are still at risk, must be discounted for that risk, and that corresponds to the risk to consumption. Importantly, it is not just that "the future is uncertain;" rather, it is a matter of accounting principle.[7] The accounting is often called historical cost accounting but is better referred to as deferral accounting. Note that fair value accounting does not honor the principle; rather, all expected future earnings are booked to the balance sheet, even though they are unrealized. However, fair value accounting has limited application, even in financial firms where it is most prominent (and which are excluded in our empirical analysis).

The relevant accounting is modeled in Penman and Zhang (2020) where it informs about the covariance term in a general, no-arbitrage pricing model. Here we convey the ideas, in terms more familiar to accountants. The accounting involved has revenue and expense components:

First, the revenue recognition principle. Revenue, the top-line number in the income statement, is booked only when uncertainty about receiving cash from customers is largely resolved—usually on making a sale with a legally-enforceable receivable booked to the balance sheet. Even then the receivable is discounted with an allowance for bad debts to reduce it to a cash equivalent (at least in principle). In asset pricing terms, revenues are recognized only when a firm can book a low-beta asset, cash or a near-cash (discounted) receivable. Until that point, revenue

[7] Accountants tend to use the word "uncertainty" while asset pricing typically refers to "risk." While there is a technical distinction that could be explored, we make no such distinction here.

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recognition is deferred as positive beta, an expectation at risk of not being realized—the customer may not materialize. Potential revenues for which the customer has performed by paying cash are not booked as revenue, but rather as a liability on the balance sheet—deferred (unearned) revenues—until the uncertainty that the firm can perform its side of the contract is resolved. Only then does the revenue go to the income statement.[8]

Second, the conservative accounting principle for booking investment to the balance sheet. Earnings is revenue minus expense, and accounting expenses also embed uncertainty. Under so-called (unconditional) conservatism, investments are not booked to the balance sheet (book value) when earnings from the investments are particularly uncertain; rather they are expensed against earnings immediately. R&D, advertising and promotion (building brand assets) are the marquee examples, but the accounting is pervasive, applied to investment to develop supply chains and distribution chains, employee training and retention, software development, start-up and organization costs, and more.[9] In contrast, investment that is booked to the balance sheet is deemed to be of lower risk; for example, in contrast to R&D investments with no evident product, inventory and plant investments are made with a saleable product in view. Conditional

[8] See the FASB's ACS 606 and IFRS 15 standards on revenue recognition where the criteria are the satisfaction of a contract on both sides and receipt of cash "highly certain." These new standards replace earlier standards, but the basic principle remains. The idea of tying the expected return to the timing of risk resolution is not foreign in asset pricing. See Epstein and Zin (1989) and Bansal and Yaron (2004), for example. Schlag, Thimme, and Weber (2020) estimate a premium for the timing of risk resolution from option data. In the accounting literature, Dechow, Sloan, and Soliman (2004) introduce the idea of equity duration tying to risk, though they have cash flow realizations rather than accrual earnings in mind.

[9] The accounting is explicitly tied to risk (or "the uncertainty of future benefits") in FASB Statement No. 2 and IAS 38 on R&D accounting and IAS 37 on the recognition of contingent assets and liabilities. Penman and Reggiani (2018) provide examples. The Conceptual Framework of the FASB and IASB introduces uncertainty in recognizing assets that can be booked to the balance sheet. Indeed, IAS 16 on property, plant, and equipment requires benefits to be "probable" for the asset to be booked.

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conservatism—impairing these booked assets upon lowered expectations but refusing to recognize anticipated gains until realized—reinforces this accounting as new information arrives.

Deferring revenue recognition to the future leads to lower current earnings, but higher future earnings. Expensing risky investment against current earnings reduces those earnings but adds expected future earnings from the investment that will now involve no amortization expense or depreciation. Higher expected earnings over current earnings is expected earnings growth, and that growth ties to priced risk because of the governing accounting principles (Penman and Zhang, 2020, Proposition 1): Revenue recognition is deferred because risk has not been resolved, and investments are expensed because of uncertainty about outcomes. Earnings will be booked *if* revenues are realized and *if* the risky investment pays off—the conditional *if* indicates risk. As priced risk, it requires a discount, given by the covariance term in equation (3). Penman and Yehuda (2019) show empirically that application of the revenue recognition principle and conservative accounting for investment conveys "discount rate news" to the market.[10]

The expected returns we estimate and the fundamental factor that we construct are based on accounting observables that are determined by this accounting and thus project earnings at risk under the accounting principles. Those observables are the current earnings and book values and the line items that sum to these income statement and balance sheet totals. To incorporate both time-t observed earnings and book value (and implicitly the line items), restate equation (3) as

[10] Of course, accounting principles can be violated in practice—the "quality of earnings" issue. However, the accounting standards on revenue recognition and booking investments to the balance sheet are quite definitive and enforced by audit. A firm or auditor violating the revenue recognition standards or inappropriately capitalizing expenses to the balance sheet are subject to severe (SEC) sanctions. Nevertheless, our empirical tests (with GAAP earnings) could be affected, though that does not seems to be the case.

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$$P_t^T = \frac{B_t}{R_T^f} + \frac{Earnings_t}{R_T^f} \sum_{\tau=t+1}^{T} E_t \left[\frac{Earnings_\tau + d_\tau (\frac{R_T^f}{R_\tau^f} - 1)}{Earnings_t} \right] + \sum_{\tau=t+1}^{T} \text{cov}_t (Earnings_\tau, m_\tau) \quad (4)$$

This brings the focus to expected (cum-dividend) earnings growth over current earnings, the growth feature that indicates priced risk in Penman and Zhang (2020) and the feature on which our empirical analysis focuses.

Appendix 1 of the Internet Appendix compares the modeling here with other models that incorporate accounting numbers, including Fama and French (2015), Voulteenaho (2002), and the Hou, Xue, and Zhang (2015).

3 Estimation of Expected Returns Conveyed by Accounting Numbers

The empirical analysis first calculates expected returns from accounting numbers that project earnings growth at risk. It then constructs a pricing factor from these expected returns. This section describes the calculation of expected returns and the rationale behind it.

3.1 The Estimation of Expected Returns

The identification of the accounting observables that project expected returns follows a preset program:

(i) Identify accounting characteristics that *a priori* (under accounting principles in Penman and Zhang, 2020) indicate expected future earnings growth at risk.

(ii) Confirm empirically that the identified characteristics do forecast future earnings growth.

(iii) Confirm that the characteristics that forecast growth also forecast returns, and in the same direction as they forecast growth. If so, they are deemed to indicate risk and the expected return associated with the growth.

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(iv) Estimate a parsimonious model that combines the characteristics into a return prediction.

(v) Estimate an out-of-sample expected return, ER_{it}, for each firm each year with coefficients estimated in sample in step (iv).

(vi) Update ER_{it} each year as fresh (annual) financial statement information arrives.

Step (i) replaces data mining with identification guided by principle. Given clean surplus accounting and a constant required return, *r*,

$$P_t^T = \frac{\sum_{\tau=t+1}^{T}[Earnings_\tau + Dividend_\tau((1+r)^{T-\tau}-1)]}{(1+r)^T-1} \rightarrow P_t,$$

the no-arbitrage price in the infinite-horizon dividend discount model, as T → ∞ (Ohlson 1995). That is, price is expected cumulative future earnings with dividends reinvested, capitalized at the required return. (The numerator here is the same as that in equations (3) and (4) but with a risk-adjusted discount rate rather than the covariance term.) Transposition renders the required return:

$$(1+r)^T - 1 = \frac{\sum_{\tau=t+1}^{T}[Earnings_\tau + Dividend_\tau((1+r)^{T-\tau}-1)]}{P_t^T} \quad (5)$$

for T → ∞. Thus, the required return is given by (long-term) expected cum-dividend earnings, discounted in the price for the risk to the expected earnings, that is, the expected earnings yield.

Accordingly, the starting point for the construction is the current earnings-to-price, $Earnings_t/P_t$ and this is the first variable to enter our estimation. If the current yield is a sufficient statistic for the expected future yield, then no further information is needed. While numerous papers have shown that earnings-to-price predicts returns, this no-growth condition is not typical, so other variables that indicate earnings growth enter. However, the other aspect is the risk to that growth, the discount to the denominating price in equation (5), so added variables must convey that risk.

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The first of these variables is book-to-price, B_t/P_t. Given E_t/P_t, adding B_t/P_t effectively adds ROE_t, and ROE captures the effect of the accounting principles via the effect of conservative accounting on both its numerator and denominator. This is developed in the Penman, Reggiani, Richardson, and Tuna (2018), Penman and Reggiani (2018), and Penman and Zhang (2021) with empirical validation. To E/P and B/P (ROE) are added other accounting variables connecting to expected growth at risk in theory. The appendix expands on the rationale.

Step (ii) validates the selection by testing the explanatory power of the variables in actually forecasting earnings growth, via the following cross-sectional regression estimated each year:

$$\frac{\Delta Earnings^{a}_{it+2} \times 2}{\left|Earnings^{a}_{it+2}\right| + \left|Earnings_{it+1}\right|} = \alpha + \beta_1 \frac{Earnings_{it}}{P_{it}} + \beta_2 \frac{B_{it}}{P_{it}} + \sum_{j=3}^{K} \beta_j A_{jt} + u_{it+2} \qquad \text{(A)}$$

Earnings and prices are per share, and $Earnings^{a}_{it+2} = Earnings_{it+2} + (r_{ft+2} \times d_{it+1})$ where d_{it+1} is dividend per share in t+1 and r_{ft+2} is the yield on the one-year T bill for year t+2. The reinvestment of dividends recognizes that dividends can be reinvested to earn more earnings, as in equations (4) and (5). The left-hand-side growth rate ranges from -2.0 to +2.0. This measure produces a growth rate that is quite close to the standard measure with positive base earnings in t+1 but accommodates the case where the base is negative as well as compressing outliers when the growth rate is on a small base. Earnings growth two years ahead is not sufficient for the multi-period growth, of course, but survivorship bias severely limits the ability of a sample of realized growth rates to represent ex ante rates. We get similar results forecasting growth for t+3, and results (in Table 1) show that out-of-sample expected growth rates based on regression A estimates align with actual earnings growth over four years ahead.

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Step (iii) serves to validate that variables that forecast growth also indicate risk and thus forecast returns (to risk), admitting only those growth predictors that are priced as such. The cross-sectional regression for the purpose, estimated each year, simply replaces the dependent variable in Regression A with forward stock returns:

$$R_{it+1} = a + b_1 \frac{Earnings_{it}}{P_{it}} + b_2 \frac{B_{it}}{P_{it}} + \sum_{j=3}^{K} b_j A_{jt} + \varepsilon_{it+1}, \qquad \text{(B)}$$

This step imposes a consistency condition: A variable that predicts higher (lower) returns must predict higher (lower) growth if it is to convey risk under accounting principles. The parsimony feature in step (iv) then reduces the inputs required (the zoo). Penman and Zhu (2014) provide detail of coefficient estimates for both Regressions A and B.

Step (v) yields expected returns estimated in real time. For each year, t, coefficients estimated with Regression B (in sample) are applied to accounting characteristics for individual firms, *i* (out of sample) to estimate an expected return, ER_{it}, going forward. The applied coefficients are rolling means of those estimated from annual cross-sectional regressions over the 10 years prior to *t* in step (iv). These ER_{it} are re-estimated annually in Step (vi).

The identified accounting characteristics may not be the best or complete set, but our aim is to demonstrate an approach rather than elicit the definitive information set. Significantly, some of the characteristics have been identified as return predictors via data mining exercises. But we stress that the identification here is on the basis of theory and accounting principle. Given this identification, it is no surprise that these variables surface robustly in repeated data analysis, including in data in non-US markets (as they do). Some variables are those nominated as "anomalies" in the literature, but now satisfy requirements (i) and (ii) before reaching step (iii).

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That is, they are now identified, not as anomalies, but as variables that connect to risk *a priori*, with added empirical confirmation that follows in this paper.

We estimate three expected return models in step (v), an ER8 model with eight variables, and two reduced models dropping some variables that are statistically significant in the ER8 model. The idea is to determine whether being "as simple as possible" works:

ER8: E/P, B/P, Accruals, Investment, Growth in NOA, ROA, EXTFIN, NSI

ER4: E/P, B/P, Accruals, Growth in NOA

ER2: E/P, B/P

See the calculation of these variables in the appendix. Model ER8 contains variables that parsimoniously fit to forward returns in step (iv) after surviving steps (ii) and (iii). It includes financing variables, EXFIN and NSI, that are not affected by the accounting principles in the last section but lever earnings growth and add risk to that growth. ER4 includes only a parsimonious set of variables affected by the accounting principles (having to do with operations rather than financing activities). ER2 is based only on the bottom-line numbers in the income statement and balance sheet (without the line-item detail in the other two models).

3.2 Data

The out-of-sample ER8 expected return is estimated from 1981 to 2012, ER4 from 1975 to 2012, and ER2 from 1966 to 2012. The periods differ because of data availability for some variables. For ER8 and ER4, the data runs from 10 years prior to these periods for in-sample estimation of regression coefficients to be applied out of sample. For all ER, the data also runs for two years after the estimation period for post-estimation validation and calculating factor returns.

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The analysis covers all U.S. firms on Compustat for any of the years and which have stock prices and returns on CRSP. Financial firms (SIC codes 6000-6999) are excluded as their accounting differs from that described earlier as relevant. So are utility firms (SIC codes 4900-4949) where the accounting is partially due to regulation. Firms were deleted for any year in which Compustat reports a missing number for book value of common equity, income before extraordinary items, common shares outstanding, or total assets. Firms with negative book value or a per-share price of less than 50 cents were also eliminated. Prices, P_t, were observed on CRSP three months after each fiscal year, by which time annual reports must be reported, by law. (To accommodate late filers, we repeated the analysis at four months after fiscal-year end.) Monthly returns are also observed on CRSP, with annual buy-and-hold returns after this date calculated as compounded monthly returns with an accommodation for non-survivors.

4 Expected Returns and their Properties

The expected return for each firm-year, ER_{it}, is estimated at step (v) of the program. The ER_{it} are updated each year from the identified accounting numbers in annual financial statements. That update applies mean coefficient estimates from estimating regression B over the prior 10 years to these numbers. This section reports on how these ER_{it} and the accounting numbers on which they are based connect to risky growth expectations, and how that risk is manifested in investment returns. The documentation is for ER8, though findings are similar for ER4 and (perhaps surprisingly) for ER2 with its limited inputs. As a reminder, these ER_{it} are out-of-sample, expected returns an investor could have estimated in real time by reading financial statements.

4.1 Expected Returns, Earnings Growth, and Accounting Characteristics

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Each year, 1981-2012, firms are ranked on their ER_{it} and formed into 10 portfolios. Panel A of Table 1 reports the mean over years of the mean ER_{it} for these portfolios, $ER_{t.}$[11] It then reports the mean forecast of earnings growth two years ahead from applying mean coefficients from estimating Regression A. Portfolios 1 and 2 aside, the ER_t are increasing in the growth forecasts: The expected return aligns with (out-of-sample) expected earnings growth.[12]

Most of the remainder of Panel A reports the accounting characteristics from which the ER_{it} are inferred, now out of sample. With the exception of the extreme ER_t associated with loss firms, E/P is similar across portfolios 4 – 9. E/P (or the P/E ratio) indicates future earnings growth, so one might expect E/P to be increasing in the forecasted growth. But that is so holding risk and the expected return constant: If the expected return increases with higher growth because it is risky, then the expected growth goes onto the expected return rather than the price, leaving E/P unchanged. That is intriguing: While expected growth increases, P/E does not because price is discounted for risk.[13] The variation in the ER extremes is also intriguing: Negative E/P are in both the high and low ER_t portfolios, so ER_{it} distinguishes loss firms with high expected growth and returns and from those with low expected growth and returns.

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[11] As firms have different fiscal-year ends, the portfolio features in this table and Table 2 do not necessarily align in calendar time, but cover all stocks. Cut-offs for the portfolio groupings were determined at each March 31 from ranking on firms' most recent ER_{it}. March 31 is three months after fiscal-year end for the majority of firms (with December 31 fiscal years). Results are similar with December 31 fiscal-year firms only.

[12] Means of yearly medians exhibit the same patterns, as does the growth variable in Regression A calculated as $\frac{\Delta Earnings^{a}_{t+2}}{|Earnings_{t+1}|}$ (with the deletion of 1% of outliers, presumably due to small denominators); the mean growth rates for portfolios 1 – 10, fitted out of sample, were -0.074, -0.011, 0.016, 0.038, 0.050, 0.062, 0.072, 0.080, 0.090, and 0.102, respectively.

[13] Put simply with a simple valuation model, $P_t = \frac{Earnings_{t+1}}{r-g}$, so E/P = $r - g$. Thus, if the required return, r, increases with expected earnings growth, g, E/P does not change. The point can also be made from equation (4): Dividing through by $Earnings_t$ to have the (trailing) P/E on the left-hand side, expected earnings growth in the second term increases with ER_{it} but that is cancelled in the valuation by an increase in the covariance term.

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The association between the other characteristics and ER_t in Panel A accord with the predictions in the appendix. Panel A also reports the mean expected return, ER_{t+1}, re-estimated at the end of year t+1 for the same firms in the portfolios at time t. They indicate that expected returns are quite persistent, though with some reversion to the mean in the extremes, more so at the lower end. Mean reversion is expected as firms become similar to the average over time but may also be due to measurement error, likely to be greater in the extremes. In unreported results, the accounting characteristics for ER_t portfolios in the table are also persistent; their t+1 values have a similar pattern over ER_t portfolios as their time-t values.

Survivorship is a concern as we move to later tables. So, the final two columns of Panel A report the percent of non-survivors due to poor performance. There is little difference over portfolios, except in the extremes. Non-survivorship due to mergers and acquisitions is similar over portfolios, though a little higher for the higher ER portfolios.

4.2 Expected Returns and Accounting Principles

Two accounting principles connect expected earnings growth to risk: the revenue realization principle and the expensing of particularly risky investment. Panel B of Table 1 reports on how these principles affect earnings at t for the ten ER portfolios.

Contemporaneous sales (revenue) growth is considerably higher for the lower ER_t portfolios: The resolution of uncertainty with sales realizations projects lower expected returns, as accounting principles suggest. The higher sales realizations yield higher operating profit margins in the table; sales growth realizations flow through to higher realized earnings. High ER_t portfolios are those where sales growth and margin growth are not (yet) realized.

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Conservative accounting for investment is also apparent in Panel B. Two such investments, R&D and advertising and promotion are increasing in ER_t; earnings for high ER portfolios are more depressed from the expensing. Correspondingly, higher ER_t portfolios exhibit higher expected future earnings growth in Panel A.[14] The denomination in revenue compares the investment to gain future revenues to revenues that the firm can currently realize: If the added investment for future revenues is high relative to what the firm can realize currently, risk is high.

R&D and advertising are only part of expensed investment. Other investment is reported in Selling, General, and Administrative expense (SG&A). Indeed, research in Kovacs (2004) and Enache and Srivastava (2018) indicate that a significant portion of SG&A is investment. Growth of SG&A, relative to revenue, is increasing in ER_t in the panel.[15]

4.3 Accounting Payoffs for Expected Returns

Panel C of Table 1 reports accounting outcomes after time, t, when ER_t is estimated. The forecast of two-year ahead earnings growth in Panel A is now compared to realized t+2 earnings growth. With the exception of portfolio 1, actual growth rates align with the forecasted rates over ER_t portfolios.[16] Further, higher growth forecasts are associated with a higher standard deviation (STD) and interdecile range (IDR) for the actual growth rates; higher ER_t portfolios with higher

[14] It is the change that is relevant, not the level, because these items are not affected by conservative accounting if there is no growth in the investment (by the cancelling error property of accounting).

[15] As further validation that ER is driven by accounting principles, we estimated expected returns with Regression B run with the four measures in Panel B, with estimated coefficients then applied out of sample each year to estimate expected returns. The mean expected returns over years for the resulting ten portfolios ranged from 0.066 for portfolio 1 to 0.249 for portfolio 10. The corresponding actual t+1 returns ranged monotonically from 0.080 to 0.228, similar to that for ER in Table 2. This further demonstrates that we are not just packaging observed anomaly variables but rather running a preset program developed from theory.

[16] The actual earnings growth rates for portfolio 1 are relatively low. The comparison of expected and actual earnings growth, 0.119 relative to 0.023, suggests that the portfolio contains firms for which expected growth contains considerable measurement error, and so for the other extreme portfolio, 10. The lower earnings over four years for portfolio 1 in the table is the lowest of all portfolios, with portfolio 2 the second lowest.

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actual growth also have higher risk of that growth deviating from expectation, with higher probability of realizations in the tails of the distribution.[17] Similarly, the four-year aggregate actual earnings yields in the table are increasing in ER_t; A higher ER_t, denominated in price, is rewarded with higher earnings denominated in the same price expectation of earnings. That accords with the starting point for the analysis in equation (5) where the expected return is given by the expected earnings yield. That higher yield is at risk, as measured by the standard deviation and IDR of these outcomes.[18]

Panel C also reports portfolio earnings betas for t+1, the sensitivity of portfolio earnings to market-wide earnings. As (aggregate) earnings add to (aggregate) book value by CSR and (aggregate) dividends that buy consumption are paid out of (aggregate) book value, market-wide earnings is a measure of the generation of aggregate consumption by firms.[19] Thus, if variation in aggregate earnings capture variation in aggregate consumption, the betas indicate the sensitivity of consumption generated by firms to aggregate consumption. They are estimated from time-series regressions of portfolio annual earnings in the forward year, t+1 on total (market-wide) earnings for that year, both relative to beginning-of-period price. Betas for earnings changes in t+1 relative to changes in market-wide earnings are also given. All betas are forward-looking, those actually experienced during the year for which ER_t conveys the expected

[17] Realized growth two years ahead are affected by realized investment one year ahead. However, results are similar for growth in residual earnings, with book value at the beginning of the year charged at the prevailing risk-free rate for the year.

[18] A similar pattern over portfolios is observed when $Earnings_t/P_t$ is subtracted, that is, with earnings growth over earnings at time, t. The results in the table can be affected by survivor bias which, in Panel A, is higher in the extreme portfolios. However, similar patterns for the standard deviation and IDR are also observed for realized one-year-ahead earnings, EPS_{t+1}/P_t, where survivorship is less of an issue.

[19] Kothari, Lewellen, and Warner (2006) report that aggregate earnings and aggregate consumption are positively correlated.

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return. Higher (lower) ER_t forecasts higher (lower) betas: ER_t predicts (systematic) fundamental risk.

In summary, Table 1 confirms that ER reflects the accounting in Penman and Zhang (2020) that conveys priced risk. Table 2 now examines whether that is so empirically.

4.4 Risk and Return to the Expected Return

Table 2 reports the realized return outcomes for the ten ER_t portfolios. Mean and median actual returns are almost monotonically increasing in ER_t and indeed are quite similar to the ER_t estimates, impressive for out-of-sample estimation. This is similar for the last half of the sample period as the first half, indicating robustness in the relationship: The analysis yields an (out-of-sample) expected return that validates against the actual returns that investors experience. It satisfies the requirement, emphasized in Black (1993), that the primary purpose of asset pricing research is to yield an expected return, not to explain the cross-section of returns.[20]

The standard deviation of actual returns increases in the high ER_t portfolios, but otherwise the differences are small. Comparison of mean returns to the standard deviation reveals that Sharpe ratios are increasing across portfolios. However, this higher return per unit of standard deviation for higher ER_t comes with higher-order risk—an increase in the range of return outcomes, higher kurtosis (more realizations in the tails), and relative skewness measures that indicate compensation on the upside for that risk. The accounting information indicates the likelihood of returns in the extremes, and it is the extremes with which investors are particularly concerned.

[20] Both the expected returns and actual returns in Table 2 are higher than one typically expects. That is presumably due to an ex post bias: the sample period was one when equity investing paid off well—the bet on growth for U.S. stocks had a favorable outcome. Penman, Zhu, and Wang (2019) calculate shrinkage estimates of the expected return to accommodate this ex post bias. For the purpose of forming a factor based on ER_t, (below), this is not an issue: ER_t portfolios can just be viewed as indicating risk classes.

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The table also reports that the ER_t predict forward return betas, complementing the earnings betas in Table 1. The return betas are increasing over ER_t portfolios, though portfolio 1 reports a high beta. Betas in up-markets are also increasing in ER_t, as are down-market betas. Significant for what follows, the portfolio 1 up-market beta is 1.77, but the down-market beta is only 0.48 and low also for portfolios 2-4. Downside beta carries a risk premium in Ang, Chen, and Xing (2006), so it appears that low ER_t provide a hedge in down markets. Not so for high ER_t where beta sensitivity to down-markets is relatively high. These betas are those experienced by investors during the t+1 year. The association of ER with these forward-looking betas indicates that the underlying accounting, though nominally dealing with firm-specific risk, actually captures beta (priced) risk. That is a requirement in Daniel, Mota, Rottke, and Santos (2020) for a characteristic to be the basis for a legitimate pricing factor rather than capturing unpriced risk.[21]

Table 2 also reports the sensitivity of ER portfolio returns to return factors that appear in the Fama and French (2015) five-factor model, HML (book-to-price), SMB (size), RMW (profitability), and CMA (investment) factors, and to returns on the I/A factor in the Hou, Xue, and Zhang (2015) (HXZ) *q*-investment CAPM. Results for the latter's size (ME) and ROE factors are similar to those for the SMB and RMW factors. The idea here is not to validate the ER_t with the betas on these factors (as if they were the "true" factors), but rather to ask if the risk associated with ER_t is exhibited by the factors, as a preamble to factor comparison tests later.

[21] We did find some evidence that contradicts the beta results. Prompted by Savor and Wilson (2014), we calculated mean daily returns for portfolios during the forward year, separating days with pre-scheduled macro news announcements from non-announcement days. There were 1,302 news days in the sample period, covering announcements of CPI and PPI revisions, employment statistics, and interest rate decisions from the FOMC. Mean returns are increasing in ER_t for the non-announcement days. However, while the mean returns during announcement days were significantly higher than non-announcement days for all portfolios, they were similar across portfolios; the higher ER_{it} firms did not earn higher average returns to compensate for risk of macro news.

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As reported in Panel A of Table 1, ER_t is positively related to B/P and here the sensitivity of portfolios returns to the HML factor increases with ER_t. (The significance of the factor betas is given in the notes to the table). While every portfolio is quite sensitive to the SMB size factor, there is little difference in the SMB betas over portfolios.[22] The betas on the profitability factor, RMW, are all negative. The betas on the investment factors, CMA and I/A, are positively related to ER_t. With investment negatively related to ER_t in Panel of Table 1 and also negatively related to investment in the *q*-investment CAPM, the alignment is agreeable—particularly as that is the prediction under accounting principles.

While our estimates are from a wide range of firm sizes, many of the existing factor models minimize the effect of micro-caps by using NYSE size breakpoints in the portfolio formation. The results in Table 2 hold over size partitions, though are stronger for the bottom 20% by size of NYSE stocks. That is expected if small firms are those with considerable growth expectations that are yet to be realized.

Internet Appendix 3 documents the time-series variation of ER_t within ER portfolios. ER_t is slowly moving over time and covaries across portfolios, indicating common sensitivity to systematic risk.

4.5 ER_{it} and the Dimensionality of Return Predictors

[22] Size, as measured by ln(market capitalization), is slightly lower for portfolio 10, 4.2 versus the typical size of about 5.4 for other portfolios. But otherwise ER_t is overall unrelated to size; accounting characteristics determine expected returns rather than size. Cochrane (2005) and Ang (2014), among others, report that the size premium in returns is not evident in periods after 1980. Penman and Reggiani (2020) show that, while ER is negatively related to size before 1980, that reverses later: High ER are more associated with large firms and these large firms have higher expected and realized earnings growth. Note that betas on the UMD momentum factor (that is in some extensions of the factor models) are consistently negative over ER portfolios.

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Research has uncovered a large number of characteristics that predict returns, but too many to suspect that they do so independently.[23] There is a need to reduce the dimensionality, not only to exclude those that just load with a "significant t-statistic" by luck, but also to produce a parsimonious set of predictors from the many correlated variables.

Lewellen (2015) does so, with out-of-sample validation, but as a statistical exercise that isolates predictors deemed "insignificant" given other included variables. In contrast, our paper reduces the dimensionality by filtering variables on the basis of organizing principles that connect them to risk. In the Lewellen paper, book-to-market, size, and momentum explain a good deal of the cross-sectional variation realized returns.

Table 3 draws a contrast. It reports results of regressions of forward realized returns, R_{t+1}, on out-of-sample ER_t, with and without variables identified as relatively important in the Lewellen paper. The regressions are estimated in the cross section each year with mean coefficients over years and t-statistics on the means (Fama-Macbeth style) reported. The regression at the portfolio level with ER_t alone produces a mean R^2 of 0.55, as is expected from the alignment of ER_t with actual t+1 returns in Table 2. The R^2 is much reduced with individual firms, of course, but the t-statistic on ER_{it} is 6.12. Remarkably, the mean coefficient of 0.979 is very close to 1.0, with a t-statistic for the mean relative to 1.0 of -0.17 (and 0.37 on the mean coefficient of 1.065 for the portfolio regressions). In both cases, the mean intercept is not significantly different from zero; the projection from ER to actual returns is with zero mean error. The slope of 1.0 takes on further significance with the understanding that it is an estimate of the return on a long-short portfolio of

[23] Harvey, Liu, and Zhu (2016) find 316 predictors, a number they say likely under-represents the total. Green, Hand, and Zhang (2013 and 2017) find that, of 333 characteristics that have been reported as predictors of stock returns, many predict returns incrementally to each other.

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left-hand-side assets with unit ER_{it}, as explained in Fama (1976): one-for-one. The estimated intercept is the return on a portfolio with zero amount of ER_{it} and reports a zero return.

The coefficient on ER_{it} changes little with the addition of the variables that are important to explaining returns in the Lewellen paper, and none of those variables loads with a significant t-statistic.[24] Of course, they may be collinear so may add explanatory power jointly. The R^2 of 0.050 compares with the 0.015 with ER_{it} alone, but the added variables are fitting in-sample (where the R^2 informs about the amount of observed variance explained), not out-of-sample (where predictive ability is to be explained).

5 The Fundamental Factor and its Properties

Fama and French (2020) explain that the slope coefficient in Table 3 can be interpreted as the return on a risk factor that expresses the stochastic discount factor, m_τ, in terms of observables. The tie of ER_{it} to consumption and to the risk to consumption cues the identification of such a factor. This section constructs that factor, reports on its properties, and compares it to extant factors. Like those factors, the constructed factor is a "factor-mimicking" portfolio formed from firm characteristics but, in this case, the characteristic, $ER_{it,}$ is explicitly identified with risk.

5.1 Construction of the Fundamental Factor

Each March 31, we construct ER8, ER4, and ER2 factors from ranking firms on their expected returns, ER_{it}, estimated from their most recent annual financial statement information. The long side of the factor is composed of the highest 30% of ER_{it} (firms in portfolios 8 - 10 in Tables 1 and 2) and the short side the lowest 30% (portfolios 1 – 3). So, the factor sells the features

[24] We also added all the variables in Panel A of Table 1 to ER_t to assess whether the ER_t captured all the information in the variables underlying its construction. Only ΔNOA_t loaded with a significant mean coefficient (with a t-statistic of -2.14).

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associated with low ER_{it} and invests the proceeds in the features associated with high ER_{it}. For December 31 fiscal-year firms the information for determining ER_{it} is almost immediate. For firms with fiscal years other than December 31, an updated ER_{it} is calculated when new financial statements arrive, resulting in a rebalancing of the long and short portfolios three months after their fiscal year if, for the cut-off points established at the prior March 31, the stock changes its portfolio position. This ensures that the factor is based on the most recent relevant information. However, results are similar when we maintain the portfolios at each March 31 for the following 12 months, avoiding rebalancing costs. This is no surprise for a slow-moving estimate with rebalancing occurring only with a migration out of or into portfolios 1 - 3 or portfolios 8 - 10.

The long and short portfolios are equally weighted, to capture the ER_t characteristic alone. Value-weighted portfolios emphasize large firms while growth and its risk are higher in small firms (Penman, Reggiani, Richardson, and Tuna, 2018 and Penman and Reggiani 2020). So value weighing shifts the focus away from the attribute of interest to another (size) where the feature is less prevalent. For example, if a large firm were exposed to expected growth at risk (as some are), sensitivity of its returns to a value-weighted ER portfolio return would not pick up the exposure. The risk in a small firm would similarly not be calibrated. Internet Appendix 2 elaborates on the point and reports results with value-weighted portfolios to compare to Tables 5, 6, and 7 below. They are not as strong as those with equally-weighted portfolios, demonstrating the point.

5.2 Comparative Properties of the Fundamental Factor

The comparison with other factors not only evaluates factors on their relative properties, but also reports which factors have similar properties to the fundamental factor and vice versa. That is

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important for factors constructed from data analysis without much reference to fundamentals. Do they capture the fundamental risk in ER_{it}?

Panel A of Table 4 reports comparative statistics summarizing monthly returns for various factors. The fundamental factors are constructed with the exclusion of microcaps (below the 20th percentile of NYSE stocks by size) to make them comparable to other factors. However, results with all stocks in the factors, ER2(ALL), ER4(ALL), and ER8(ALL) in the table, are included.

The ER8 factor yields a mean return of 0.571% per month, among the highest for all factors. It is also higher than the means for the ER2 and ER4 factors, the out-of-sample improvement from the added variables in ER8.[25] The ER8 mean relative to the standard deviation, in the t-value of 4.07 and the annualized Sharpe Ratios of 0.720, are also among the highest. The BAB mean return stands out as particularly high at 0.909%, though this factor is constructed for U.S. stocks over the full-size range, as is the quality factor (QMJ) with a mean 0.578%. So, the appropriate comparison is to the ER8(ALL) number, 0.790% with its Sharpe Ratio of 1.157. The kurtosis for ER8 is also high, identifying higher tail risk.[26]

Panel B of Table 4 reports correlations between the monthly returns on the various factors. The ER factors are negatively correlated with the excess market return (MKT-R_f), more so for the ER(ALL) factors. The ER factors are strongly positively correlated with HML, CMA, and I/A, as Table 2 indicates (and for reasons explained there). The correlations of the ER factors with BAB

[25] The range of mean monthly returns (in percent) for ER2 portfolio 1 and 10 is 0.540 to 1.220, 0.239 to 1.411 for ER4, and 0.176 to 1.559 for ER8, an increasing spread with more variables. The mean in-sample adjusted R^2 from estimating Regressions A and B for ER2, ER4, and ER8 are respectively 0.027, 0.030, and 0.033 for Regression A and 0.024, 0.030, 0.041 for Regression B.

[26] Sharpe Ratios are annualized (approximately) by multiplying the monthly ratios by sqrt(12). We repeated the analysis here and in Panel B with the annual returns in Table 2. The annual Sharpe ratios for ER2, ER4, and ER8 are 0.512, 0.705, 0.719, respectively, compared with 0.376 for HML, 0.117 for SMB, 0.398 for RMW, 0.532 for CMA, 0.740 for I/A, and 0.595 for ROE.

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and QMJ are not high, though more so when, like BAB and QMJ, the ER factor is constructed from all firms. The three ER factors are strongly correlated with each other.[27]

5.2. Time-series Tests Against Other Factors

Table 5 examines the extent to which ER8 factor returns are explained by other factors, so-called spanning tests. The intercepts (alphas) test whether the ER factor yields average returns not captured by the comparison factors (their pricing errors). Estimated slopes indicate commonality with those factors.

The first regression reports that ER factor returns have little correlation with the market, and the ER factor yields a high alpha intercept: These two portfolios are very different. The ER factor loads on several of the other factors, particularly the book-to-price HML factor and the investment factor, I/A, which capture B/P and investment that enter the ER8 factor. But, in all cases, we observe significant alphas, against the factors in the Fama and French three-factor model, the Fama and French three-factor model with an added momentum factor (UMD), the Fama and French five-factor model, and the *q*-investment CAPM. These models explain at most half the variation in the ER factor returns. Returns on the fundamental factor are not explained much by BAB and QMJ factors. When all factors are included in the last column, the R^2 increase to 61.3 %, but the estimated intercept is 0.366% per month with a t-statistic of 3.01.

Table 6 reverses the regressions, now regressing returns for factors in Table 5 on ER8 factor returns. The intercept of 0.825% in the regression of MKT – R_f on ER8 is large: Again, these two portfolios are very different. The intercepts for many of the other factors are not significantly different from zero, with the exception on the UMD momentum and ROE factors. With a low R^2,

[27] Legitimate factor returns cannot be highly serially correlated. The first order serial correlation of ER8 returns is -0.082.

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BAB and QMJ appear to be driven differently from ER8 (and results are similar for the ER8(ALL) factor). (Note that the t-statistics here involve multiple comparisons.)

The UMD finding is interesting, for fundamental investors claim that momentum is pricing that departs from fundamentals. Returns to QMJ and BAB have been interpreted as behaviorally driven; investors ignore stable quality stocks in favor of firms with more glamorous prospects and, with constraints on borrowing, favor high beta stocks, affecting their pricing. These are just conjectures, of course, for these factors are not well understood. The ROE factor is from a production-investment (supply-side) model rather than a consumption (demand-side) model, thus explicitly does not capture consumption at risk. However, the result here raises the question that such a model captures risk and return not captured by a consumption-based model.

6 A Two-Factor Model?

The fundamental factor yields a positive mean return with zero net investment, and with negative correlation with the CRSP market index return. It thus has the appearance of a portfolio that yields return for risk, but risk that is not related to beta on the market portfolio. What could this risk be? Two interpretations are on the table.

6.1 Better Betas?

First, following the Roll (1977) critique, the CRSP value-weighted index used here is not highly correlated with the "true" market portfolio that investors hold as mean-variance efficient. Thus, the Security Market Line (SML) from estimated betas is flatter than that which investors perceive. ER8 then corrects to the appropriate steeper line, indicating returns associated with "true" low beta and high beta. In the Hansen and Richard (1987) embellishment of the Roll

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(1977) point, the true market portfolio is identified by information that investors use but the econometrician does not. We have possibly identified that information.

The SML is usually estimated with historical betas. Table 2 reports that ER_t predicts forward, experiential betas, though these are calculated against the CRSP index, not the "market portfolio." More forcefully, Panel B of Table 1 shows that ER_t is positively correlated with forward earnings betas. As investors are buying future earnings that buy consumption, it is the shock to earnings that defines their risk with respect to the market portfolio. So, the accounting information could be conveying forward-looking "better betas" than historical betas.

6.2 A Hedge Portfolio?

Table 2 suggests a second interpretation. The first interpretation is within a single-period CAPM, but investing over many periods involves dealing with uncertainty about state variables that project consumption in the future. The high ER_t portfolios in Table 2 have relatively high beta, both in up- and down-markets. However, while the low ER_t portfolio experiences a high upside beta, the down-side beta is relatively low, also for (low) portfolios 2 and 3: These portfolios provide protection in down markets. Could these be portfolios that provide a hedge against loss of future consumption when aggregate consumption and wealth are down and marginal utility is high, as in the Merton (1973) Intertemporal CAPM (ICAPM) and Campbell (1993, 1996)?

The formal link of the fundamental factor to the risk to consumption promotes this interpretation. In Panel A of Table 1, low ER_t portfolios are associated with accounting realizations that indicate an ability to deliver consumption—sales growth, profit margin growth, higher accrued earnings realizations, investment growth, growth in NOA and financing for new projects. These are features that suggest that the firm is safer for consumption, less affected by

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changes in state variables. Following the reasoning in Heaton and Lucas (2000) and Santos and Veronesi (2006), they bode well for the risk to returns from human capital, for those returns come from firms that employ humans and provide pensions that must be securely funded to yield future consumption; working for these firms is a safer place. Growth in NOA (book value from operating activities) is particularly significant: Because dividends that buy consumption are paid out of book value, growth in realized book value indicates the ability to pay those dividends despite state outcomes.

If the low ER_t portfolios provide a hedge to consumption risk, then, by selling low ER_{it} stocks and investing the proceeds in high ER_{it} stocks—buying the fundamental factor—the investor (speculator?) is surrendering the hedge and investing in riskier stocks. The fundamental factor portfolio should thus earn a positive mean return as compensation, as it does.[28]

The financial crisis of 2008-2009 provides an opportunity to investigate. That was a time when wealth and consumption growth were damaged, and future consumption looked very much at risk. Table 7 reports monthly market returns from September 2008, when the crisis hit, through December 2009. Conveniently, this period can be split into two, one with large drawdowns for the market, September 2008 to February 2009, and one of recovery, March 2009 to December 2009. This permits an examination of the performance of ER8 under two conditions, (large) negative shocks to wealth and (large) positive shocks. So, along, with the factor return, the table reports the monthly returns on the long side and short sides of ER8(ALL), ER-H and ER-L, with the totals for the monthly returns during the two conditions also given. In the down-state, ER-H

[28] The BAB factor has been interpreted as indicating a steeper SML, with leverage and borrowing constraints entering the argument. While the correlation of the ER8(ALL) factor with the BAB factor in positive (0.18 in Table 5), it is not high. Further, both the long and short side of ER8(ALL) have low correlation with BAB, -0.11 and -0.17, respectively. On the face of it, then, the second interpretation from the inter-temporal asset pricing model is more palatable. It is unclear that the BAB return spread holds under a conditional CAPM (Cederburg and O'Doherty 2016).

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performs considerably worse than the market, but the ER8-L return is higher. With the insurance from ER-L, the total of monthly returns earned by the factor was -7.48% in this down period relative to -50.54% for the market. In the up-state, ER-H performs very well, but ER-L less so, rewarding the investor for foregoing the safer stocks and investing in the risker ER-H. The ER8 factor (long minus short) earned 35.95% relative to the up-market return of 45.71%, the cost of insurance. In sum, the short-side of the factor performs relatively well in conditions when risk to future consumption is high—protection in bad times—while the long side performs relatively well with the improving economic outlook.

The findings from the financial crisis generalize. The left-hand side of Table 8 reports the results of time-series regressions of excess monthly returns for ER_t portfolios on the excess return on the market (CRSP index) and the ER8 factor. In contrast to Tables 1 and 2, the ER_t portfolios are formed excluding microcaps, as with ER8. But results are similar with portfolios formed with all stocks and the ER8(ALL) factor. The mean monthly excess returns increase over ER_t deciles, corresponding to the spread of returns in Table 2. However, the alphas are insignificantly different from zero, indicating returns are explained by the two factors (and with considerable R^2), though the intercept t-statistic for portfolio 1 is marginally significant. The β_1 coefficient on the market factor is similar across portfolios, a little higher in the extreme ER_t portfolios. These betas largely cancel in the long-short ER8 portfolio; the difference in average beta for portfolios 8-10 and portfolios 1-3 is only 0.078. However, β_2 on ER8 is increasing in ER_t. As ER8 involves selling the hedge in the low ER_t portfolios and exposing investment to risk without the hedge, the positive β_2 for high ER_t portfolios indicates that these portfolios add expected return over the market portfolio for this exposure. The lower returns for the low ER_t portfolios, in comparison, are attributable to negative β_2; buying the hedge against bad times

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means a lower expected return over all conditions; the lower return for these portfolios is the cost of protecting consumption (insurance). Of course, for portfolios 1-3 and 8-10, the return for the ER_t portfolio is part the ER8 factor, ER8-H and ER8-L respectively, yielding $\beta_2 = 0$ if ER8-H and ER8-L returns were perfectly correlated (orthogonal to the market return). The $\beta_2 \neq 0$ result indicates that, given the market return, these two portfolios' returns vary differently.

The right-hand side of Table 8 regresses the excess return for ER_t on the excess market return and that for the short side of the ER8 factor, ER8-L, the portfolio that provides the conjectured hedge. (Portfolios 1 - 3 make up ER8-L, so are omitted.). With the market return in the regression, the β_2 exposure is exposure to the orthogonal component to the market return, a portfolio hedging market risk. Both β_1 and β_2 are significantly positive, with high R^2 for all portfolios. The beta on the market return, β_1, is reduced; buying the hedge cuts out some of the upside and downside from market volatility but buys sensitivity to the hedge. The weights on the market and ER8-L, $\beta_1 + \beta_2$, approximately sum to 1.0 for all ER_t portfolios; all portfolios involve allocating the investment dollar to the market portfolio and the hedge portfolio. β_1 and β_2 vary little over ER_t portfolios; investment dollars allocated to the market portfolio and hedge portfolio with these weights yield approximately the same return for every portfolio, with differences in mean return given by the intercept, the reward for additional risk associated with ER_t.

These results point to a two-factor model that "looks like" the Merton intertemporal ICAPM that includes the market factor but with added factors that correlate with changes in state variables. Here, however, these multiple factors are captured in one asset that satisfies a hedging demand and a reverse hedging demand from speculators. As the model is consumption based, it also resonates with the Breeden (1979) consumption CAPM which also incorporates exposure to

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state variables but with all exposures (including that to the market portfolio) incorporated in sensitivity to one factor, aggregate consumption, rather than two. With aggregate consumption hard to measure, the two-factor representation appears to be more operational.

The consistency in the weights across portfolios in Table 8 invites the question of whether this is observed in portfolios more generally. Internet Appendix 4 reports on the same tests as those in Table 8, but for the 25 Fama and French portfolios that interact size (ME) and book-to-market (BE/ME). Results are similar: Returns explained by the two-factor model with a similar weighting of the two factors. A follow-up paper, Penman, Zhu, and Wang (2021), finds that the two-factor model explain returns on a wide range of risky portfolios, including industry portfolios, those underlying extant factors, and returns associated with so-called anomalies, such as "value," "growth," size, and P/E. The exception is momentum returns, often attributed to mispricing, raising the specter of the model identifying mispricing.

7 Conclusion

The paper develops an accounting-based asset pricing model that recasts the consumption pricing model in terms of accounting outcomes that tie to consumption and the risk to consumption under accounting principles. The model yields estimates of expected returns based on accounting fundamentals that forecast realized returns and the risk to those returns. Correspondingly, the expected returns forecast earnings that buy consumption and the risk to those expected earnings (and consumption). It also leads to the construction of a fundamental pricing factor. This factor explains returns not captured by extant factor models, but also serves as a benchmark for evaluating whether those factors capture risk and return conveyed by fundamentals. The factor has negative correlation with the market return and provides a hedge against bad times when the market is down and consumption is presumably low. Exposure to the zero-net-investment factor

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yields a positive mean return, interpreted as a reward for the risk from shorting the hedge portfolio and investing the proceeds in stocks with no hedge. The insight leads to a two-factor model with the market portfolio and a hedging portfolio that corresponds to a two-factor consumption ICAPM and which validates in the data.

Underlying the analysis are accounting principles that recognize risk, principles that inform about the asset pricing discount factor as a matter of theory. Determined by these accounting principles, observed accounting numbers convey information about risk and expected return. This is exploited to yield expected returns and to construct the fundamental factor.

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Appendix. Identification and Calculation of Accounting Variables

This appendix lists the accounting variables, $\frac{E_t}{P_t}, \frac{B_t}{P_t}$ and $A_j, j = 3,\ldots, K$, that enter as predictors of earnings growth and returns in Regressions A and B. The variables are accompanied by an explanation of why they are prompted by accounting principles that tie expected earnings growth to risk in Penman and Zhang (2020) and thus enter under step (i) of the program.

The identification of earnings-to-price (E/P) and book-to-price (B/P) is explained in Section 3.1 (and further below). Additional accounting variables enter if they forecast growth in Regression A and survive in Regression B, that is, they satisfy the validation requirements (ii) and (iii), as well as being identified by accounting principle. Where support for the variable selection is provided by extant accounting and asset pricing research, it is noted. Summary statistics for variables and in-sample estimation of regressions A and B are in Penman and Zhu (2014).

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Variable	Calculation	Explanation for Selection	Predicted and Actual Sign of Coefficient in Regression A	Predicted and Actual Sign of Coefficient in Regression B
Earnings-to-price (E_t/P_t)	Reported earnings-to-price for year t, calculated as earnings before extraordinary items (Compustat item IB) and special items (item SPI), minus preferred dividends (item DVP), with a tax allocation to special items at the prevailing Federal statutory corporate income tax rate for the year. Earnings and prices are on a per-share basis, with prices observed three months after fiscal-year end adjusted for stock splits and stock dividends during the three months after fiscal year end.	E/P is the starting point, for E/P indicates the expected earnings yield in equation (5) when there is no anticipated earnings growth. Variables (below) are then added if they forecast earnings growth at risk.	Positive	Positive
Book-to-price (B_t/P_t)	Common equity at the end of fiscal-year t, divided by price at t. Book value is Compustat's common equity (item CEQ) plus any preferred treasury stock (item TSTKP) less any preferred dividends in arrears (item DVPA). Book value and prices are on a per-share basis, with prices at three months after fiscal-year end adjusted for stock splits and stock dividends during the three months after fiscal-year end.	For a given E/P, B/P recovers earnings-to-book (ROE) that captures the effect of earnings deferral and conservative accounting on earnings and book value: ROE: ROE = E/P × P/B (approximately). See Penman and Zhang (2021) for the explanation and for empirical documentation tying ROE to risky growth expectations and to returns. Penman and Reggiani (2018) provide examples. Penman, Reggiani, Richardson, and Tuna (2018) document that, given E/P, B/P (and implicitly ROE) forecasts earnings growth and the risk surrounding the expected growth.	Positive	Positive
Sales growth rate$_t$	$\frac{\Delta Sales_t}{Sales_{t-1}}$	Sales (revenues) are a realization of deferred revenue expectations, resolving uncertainty (and reducing risk) under accounting principles. Correspondingly, unrealized (expected) sales (with low realized sales growth) indicate higher risk (that the expectation might not be realized). Penman and Yehuda (2019,	Negative	Negative

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		Table 7) show that sales realizations convey (lower) discount rate news.		
Change in operating profit margin$_t$	Change in earnings before net interest expense and taxes divided by sales	This further captures the realization principle, the increase in realized operating profit from recognized sales. That realization indicates lower risk under accounting principles.	Negative	Negative
Return on Assets, ROA$_t$	Income before extraordinary items (IB) and net interest divided by lagged assets	The measure is the same as ROE$_t$, but unlevered and with a different denominator that captures more precisely the effect on conservative accounting on assets.	Negative	Predicted negative but not significant (with E/P and B/P which capture ROE in the regression)
Accruals$_t$	Sum of change in accounts receivable (Compustat item RECT), change in inventory (item INVT), and change in other current assets (item ACO), minus the sum of change in accounts payable (item AP) and change in other current liabilities (item LCO), minus depreciation and amortization expense (item DP), all divided by average assets	Accruals are driven primarily by sales recognition (receivables net of revenue deferrals) from realization. But they also include associated expense accruals that allocate costs to the current period versus deferral to the future. Lewellen and Resutek (2019) report that higher accruals are associated with lower future earnings (growth).	Negative	Negative
Investment$_t$	Change in gross property, plant, and equipment (item PPEGT) + change in inventory (item INVT), all divided by lagged assets	Investment booked to the balance sheet is the realization of (uncertain) investment opportunities and thus a revision of the risk that those opportunities will be realized. In making these investments, the firm signals that earnings can be realized, and realized earnings imply lower risk under accounting principles. This is a restatement in accounting terms of the connection between the exercise of real options and expected returns in Berk, Green, and Naik (1999): Risk declines as firms convert growth options to "assets in place." Further, under conservative accounting, investment booked to the balance sheet is deemed lower risk for earnings outcomes than investment that is expensed immediately. Oh and Penman (2021) find the market discounts investment expensed to income statement	Negative	Negative

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		relative to that booked to the balance sheet. The conservative accounting treatment for particularly risky investments corresponds to the real options view where investments (such as R&D) pertain to future growth options rather than assets in place from realizing investment options. The connection of investment to risk is also made in the Cochrane (1991) *q*-theory under which firms make investments when the hurdle rate for investment is lower (and the prospect of realizable earnings lowers that rate). The Merton (1973) intertemporal CAPM and the Liu, Whited, and Zhang (2009) Investment-CAPM provide formal links of investment to expected returns. The Conceptual Framework of the FASB and IASB introduces uncertainty in prescribing assets that can be booked to the balance sheet. IAS 16 on property, plant, and equipment requires benefits to be "probable" for the asset to be booked. IAS 38 and FASB 2 require investments to be expensed if outcomes are particularly uncertain.		
Growth in net operating assets (ΔNOA_t)	Change in net operating assets divided by average assets. Net operating assets is the sum of accounts receivable (item RECT), inventory (item INVT), other current assets (item ACO), property, plant, and equipment (item PPENB), intangible assets (item INTAN) and other long-term assets (item AP), minus the sum of accounts payables (item AP), other current liabilities (item LCO) and other long term liabilities (item LO)	This is the change in total net operating assets (operating assets minus operating liabilities) due to realized income from operations and realized investments. It includes recognized earnings not captured by the accruals measure, for example, income in subsidiaries, realized gains and losses on assets sales, long-term accruals such as deferred taxes and deferred revenues, and impairments and restructurings. ΔNOA is lower if (relatively risky) investments are expensed with conservative accounting. Ohlson and Chu (2019) report that ΔNOA forecasts future sales and earnings realizations—it projects the ability of the firm to realize earnings.	Negative	Negative
External financing ($EXTFIN_t$)	Change in debt plus the change in equity from net equity transactions, scaled by average assets. Change in debt is the cash proceeds from the issuance of long term debt (item DLTIS) less cash payments for long	External financing is positively correlated with current investment, and also indicates plans for further investment. Lamont (2000) shows that investment plans are negatively correlated with future returns. These investments and investment plans are		

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	term debt reductions (item DLTR) plus the net changes in current debt (item DLCCH). Change in equity is measured as the proceeds from the sale of common and preferred stock (item SSTK) less cash payments for the purchase of common and preferred stock (item PRSKC) less cash payments for dividends (item CDVC).	realizations (or anticipated realizations) of uncertain investment opportunities which, in turn, signal that expected sales and earnings can be realized in the future (as above).	Negative	Negative
Net share issue (NSI_t)	The natural log of the ratio of split-adjusted shares outstanding at the end of the fiscal year to shares outstanding at the end of the previous fiscal year.	For given total financing (EXTFIN), net share issues reduce future earnings per share (growth) and reduce leverage (risk).	Negative	Negative
Forward returns, R_{t+1}	Annual buy-and-hold annual returns beginning three months after fiscal-year end, the point at which ER_{it} is estimated. The annual returns are compounded monthly CRSP returns. For firms delisted during the return period, the remaining return for the period was calculated by first applying CRSP's delisting return and then reinvesting any remaining proceeds in the size-matched portfolio (where size is measured as market capitalization at the start of the return accumulation period). This mitigates concerns about potential survivorship bias. Firms that are delisted for poor performance (CRSP delisting codes 500 and 520–584) frequently have missing delisting returns; a delisting return of -100% is applied in such cases.	Actual forward reward returns, the investor's one-year return from investing based on estimated expected returns, serve as the validation of the expected return estimate.	NA	NA

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Table 1

Accounting Characteristics for Expected Return Portfolios

The table reports mean expected annual returns, ER_{it}, and associated accounting characteristics for ten portfolios formed on the ER_{it} estimated out of sample. Reported numbers are means over years of means for the ten portfolios formed each year, 1981-2012. Panel A and B characteristics are those at the time, t when ER_{it} is estimated, and Panel C outcomes are those observed subsequently. The characteristics in Panel A are those out of sample that were found to project earnings growth and returns parsimoniously in sample. Those in Panel B pertain to accounting principles that govern earnings measurement and affect earnings growth and convey risk. ER_{it} is estimated by estimating Regression B in the cross-section with variables identified in Regression A, and then applying mean coefficients estimates over the prior 10 years to accounting characteristics in each year on a rolling basis. Forecasts of EPS growth rates two years ahead are estimated by applying mean coefficients from estimating Regression A over the prior 10 years to out-of-sample accounting characteristics in each year. The appendix describes the calculation of variables.

Panel A: Accounting Characteristics at Time t for ER Portfolios

Expected Return Portfolio	Expected Return (ER_t)	Forecast of EPS Growth Rate Two Years Ahead	E_t/P_t	B_t/P_t	$Accruals_t$	$Investment_t$	ΔNOA_t	$EXTFIN_t$	NSI_t	ER_{t+1}	Non-survivor Rate $_{t+1}$	Non-survivor Rate $_{t+2}$
1 (Low)	0.043	0.119	-0.103	0.437	0.005	0.187	0.190	0.276	0.151	0.117	2.44%	7.09%
2	0.117	0.043	-0.015	0.471	-0.001	0.141	0.154	0.104	0.080	0.145	1.50%	3.76%
3	0.145	0.030	0.009	0.486	-0.011	0.112	0.124	0.055	0.047	0.159	1.19%	3.04%
4	0.162	0.028	0.020	0.515	-0.020	0.088	0.098	0.028	0.028	0.168	0.99%	2.58%
5	0.174	0.031	0.030	0.542	-0.030	0.074	0.076	0.011	0.018	0.175	0.95%	2.27%
6	0.186	0.041	0.029	0.610	-0.040	0.059	0.057	-0.001	0.013	0.182	1.19%	2.67%
7	0.197	0.053	0.030	0.667	-0.049	0.051	0.042	-0.011	0.007	0.187	1.04%	2.42%
8	0.209	0.070	0.029	0.775	-0.056	0.037	0.022	-0.019	0.004	0.196	1.12%	2.61%
9	0.225	0.104	0.018	0.944	-0.066	0.022	0.004	-0.026	0.000	0.205	1.55%	3.43%
10 (High)	0.264	0.215	-0.044	1.354	-0.093	-0.012	-0.043	-0.044	-0.004	0.222	3.10%	6.31%

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Panel B: Effect of Accounting Principles on Earnings at Time t for ER Portfolios

		Realization		Conservative Investment Expensing	
Expected Return Portfolio	**Expected Return (ER_t)**	**Sales Growth Rate$_t$**	**Change in Operating Profit Margin$_t$**	**Change in R&D + Advert to Revenue**	**Change in SG&A to Revenue**
1 (Low)	0.043	0.374	0.037	-0.021	-0.015
2	0.117	0.255	0.024	-0.010	-0.006
3	0.145	0.195	0.009	-0.003	-0.004
4	0.162	0.145	0.007	-0.003	-0.003
5	0.174	0.121	0.005	-0.004	0.000
6	0.186	0.097	0.003	0.002	0.000
7	0.197	0.078	0.001	0.000	0.001
8	0.209	0.055	0.001	0.000	0.002
9	0.225	0.035	0.005	0.000	0.005
10 (High)	0.264	-0.001	0.011	0.003	0.012

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Panel C: Accounting Outcomes Subsequent to Time t for ER Portfolios

Expected Return Portfolio	EPS Growth Rate Two Years Ahead				EPS Over Next Four Years/P_t				
	Forecast	Actual	STD Actual	IDR Actual	Actual	STD Actual	IDR Actual	Earnings $Beta_{t+1}$	Earnings Change $Beta_{t+1}$
1 (Low)	0.119	0.023	0.154	0.361	0.013	0.072	0.192	0.45	0.84
2	0.043	0.008	0.157	0.459	0.091	0.048	0.112	0.71	0.81
3	0.030	0.012	0.163	0.425	0.134	0.050	0.098	0.72	0.94
4	0.028	0.019	0.147	0.382	0.154	0.049	0.115	0.85	0.69
5	0.031	0.044	0.147	0.430	0.184	0.052	0.109	0.85	0.47
6	0.041	0.031	0.176	0.462	0.188	0.052	0.066	0.90	0.72
7	0.053	0.053	0.168	0.450	0.206	0.054	0.091	1.16	1.27
8	0.070	0.051	0.174	0.376	0.203	0.060	0.138	1.30	1.82
9	0.104	0.082	0.206	0.502	0.200	0.081	0.218	2.10	2.24
10 (High)	0.215	0.123	0.263	0.672	0.162	0.124	0.308	4.42	2.46

Actual earnings growth rates are calculated in the same way as forecasted growth rates, the dependent variable in Regression A. Variables are defined in the appendix. ER_{t+1} is the mean expected return at the end of year t+1 estimated for firms in the respective ER_t portfolios at time t. R&D is research and development and SG&A is selling, general, and administrative expense. In Panel C, STD is standard deviation and IDR is the interdecile range of annual portfolio means over years. EPS Over Next Four Years/P_t is the sum of split-adjusted EPS for years t+1 to t+4 with dividends for year t+1 to t+3 reinvested as the prevailing risk-free rate for the year, all divided by price per share at t, as in equation (4) for T = 4. Earnings beta is the slope coefficient from estimating the following time-series regression of portfolio annual earnings yield on the market-wide earnings yield, for December 31 fiscal-year firms only (to align in calendar time):

$\text{Portfolio}\frac{Earnings_{t+1}}{P_t} = \alpha + \beta \cdot \text{Market}\frac{Earnings_{t+1}}{P_t} + \varepsilon_t$. The portfolio earnings yield is aggregate earnings for the portfolio relative to aggregate price and the market earnings yield is aggregate earnings for all stocks in the sample for the relevant year relative to aggregate price. Earnings change betas are similarly estimated. Non-survivor rates in Panel A are those for poor performance.

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Table 2

Expected Returns, Actual Returns, and Forward Betas

The table summarizes the distribution of actual (realized) annual returns for ER_t portfolios over the year after ER_{it} is estimated. It also reports betas for these actual returns with respect to the market return (the return for the value-weighted CRSP index) and their sensitivities to factors in the Fama and French (FF) five-factor model and I/A factor in the Hou, Xue, and Zhang (2015) (HXZ) *q*-investment CAPM.

Expected Return Portfolio	Expected Return (ER_t)	Actual Annual Returns t+1						Market Beta t+1 (Annual Returns)			FF Betas t+1 (Monthly Returns)				HXZ Beta t+1
		Mean	Median	STD	Range	Skewness	Kurtosis	All Years	Up Beta	Down Beta	HML	SMB	RMW	CMA	I/A
1 (Low)	0.043	0.049	0.028	0.258	0.977	0.495	2.364	1.31	1.77	0.48	0.07	0.84	-0.31	-0.34	-0.46
2	0.117	0.094	0.042	0.211	0.847	0.074	2.344	1.13	1.35	0.81	0.10	0.78	-0.15	-0.15	-0.29
3	0.145	0.129	0.117	0.216	0.967	0.097	2.886	1.15	1.44	0.87	0.09	0.78	-0.13	-0.08	-0.15
4	0.162	0.144	0.139	0.214	1.037	0.557	3.562	1.12	1.38	0.85	0.06	0.70	-0.11	0.01	-0.04
5	0.174	0.155	0.113	0.215	1.041	0.497	3.330	1.20	1.58	0.96	0.07	0.75	-0.03	0.06	0.02
6	0.186	0.160	0.151	0.197	0.941	0.485	3.784	1.18	1.62	0.99	0.19	0.73	-0.05	0.02	0.10
7	0.197	0.179	0.174	0.201	0.990	0.337	3.569	1.20	1.65	1.12	0.17	0.70	-0.09	0.06	0.13
8	0.209	0.191	0.174	0.235	1.074	0.717	4.316	1.30	1.77	1.27	0.19	0.72	-0.14	0.04	0.22
9	0.225	0.212	0.219	0.258	1.413	1.142	6.050	1.41	2.02	1.44	0.27	0.73	-0.32	0.01	0.26
10 (High)	0.264	0.282	0.241	0.360	1.826	1.265	5.782	1.82	2.89	1.45	0.14	0.80	-0.62	0.13	0.28

Actual return metrics summarize the times series of portfolio returns in year t+1, 1982-2013. Kurtosis is relative to 3 for the normal distribution. Market betas and factor betas are estimated from the time series of portfolios returns, the market betas with annual returns and the factor betas with monthly returns. Returns are equally weighted in ER_t portfolios. All betas are estimated with December 31 fiscal-year firms only, to align returns in calendar time. Up markets are those where the CRSP value-weighted index was greater than 10% for the year, and down markets are those where it was less than -10%. FF and HXZ betas are estimated with all factors in the model (not one at a time). HML, SMB, RMW, and CMA monthly returns are those for the Fama and French book-to-price, size, profitability, and investment factors, respectively, and are from the Kenneth French website library. *q*-investment CAPM factor returns were supplied by Lu Zhang. With the exception of portfolio 1, none of the estimated betas on the CMA factor are significantly different from zero, nor are those for HML in portfolios 1-5 and RMW in portfolios 5 and 6. The t-statistics on the SML size factor are significant for all portfolios, as are those for the I/A

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factor with the exception of portfolios 4 and 5. Betas on other *q*-investment CAPM factors (size and ROE) were similar across portfolios to the corresponding factors in the FF model.

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Table 3

Coefficient Estimates from Annual Cross-sectional Regressions of Forward Actual Annual Returns (R_{t+1}) on ER_t and Other Time-t Variables

The table reports mean coefficients from estimating cross-sectional regressions each year, 1981-2012. The t-statistics (in parenthesis) are estimated from the time-series of estimated coefficients with a Newey-West correction for the serial correlation in the coefficient estimates. Size is log market capitalization of equity. Beta is the return beta on the value-weighted CRSP index return. Leverage is net debt/market value of equity. Momentum is the buy-and-hold return over the twelve months prior to one month before time t when ER_{it} is estimated.

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$$Return_{it+1} = \alpha + \beta ER_{it} + \cdots + e_{it}$$

		Individual Firm ER_{it}	
	Portfolio ER_t	**ER_{it} Alone**	**Adding Factors from Pricing Models**
Intercept	-0.034	-0.012	0.057
	(-0.80)	(-0.31)	(1.02)
ER_t	1.065	0.979	0.911
	(6.06)	(6.12)	(9.24)
E_t/P_t			-0.017
			(-0.25)
B_t/P_t			-0.012
			(-0.53)
$Beta_t$			0.002
			(0.27)
$Size_t$			-0.010
			(-1.86)
$Leverage_t$			-0.004
			(-0.66)
$Momentum_t$			-0.008
			(-0.33)
Adj. R^2	0.550	0.015	0.050

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Table 4

Summary Statistics for Factor Returns

Panel A summarizes the distributions of monthly returns for the ER2, ER4, and ER8 factors, along with those of the Fama and French (2015) five-factor model and the Hou, Xue, and Zhang (2015) *q-i*nvestment CAPM. Most of these factors are defined in Table 2, but to these have been added the ROE factor in the *q*-investment CAPM, the Betting against Beta (BAB) factor in Frazzini and Pedersen (2014) and the quality (QMJ) factor in Asness, Frazzini, and Pedersen (2019). MKT-R_f is the excess return on the market over the risk-free return. Returns for the Fama and French factors are from the Kenneth French website library, the *q*-investment CAPM factors were supplied by Lu Zhang, and the BAB and QMJ factors are from the AQR website. The period runs from April 1982 to March 2014 (384 months). To ensure appropriate comparisons to the other factors, the ER2, ER4, and ER8 factors were constructed excluding micro-cap firms, that is, those below the 20th size percentile for NYSE stocks. ER2(ALL), ER4(ALL), and ER8(ALL) factors include all stocks. Panel B reports Pearson correlations between factor returns.

Panel A. Summary Statistics for Monthly Percentage Factor Returns

Factor	Mean	Std Dev	t-value	Annual Sharpe Ratio	Min	p5	p10	p50	p90	p95	Max	Kurtosis	Skew
MKT-R_f	0.703	4.506	3.06	0.541	-23.240	-7.260	-4.780	1.170	6.080	7.190	12.470	2.578	-0.789
HML	0.313	2.944	2.08	0.368	-11.250	-3.910	-2.890	0.170	3.780	5.440	12.910	2.482	0.099
SMB	0.131	2.946	0.87	0.154	-15.280	-4.070	-3.290	0.000	3.430	4.640	18.730	5.709	0.478
RMW	0.389	2.556	2.98	0.527	-19.110	-3.050	-1.850	0.410	2.940	3.880	13.520	12.786	-0.513
CMA	0.345	2.025	3.34	0.590	-6.880	-2.540	-1.800	0.195	2.900	3.310	9.550	2.183	0.422
UMD	0.579	4.562	2.49	0.440	-34.580	-6.800	-4.710	0.685	4.880	6.540	18.380	12.447	-1.651
I/A	0.396	1.941	4.00	0.706	-7.160	-2.490	-1.880	0.365	2.690	3.350	9.250	1.859	0.180
ROE	0.558	2.621	4.17	0.737	-13.850	-3.670	-2.050	0.655	3.260	4.020	10.380	5.100	-0.773
BAB	0.909	3.708	4.80	0.849	-15.132	-6.022	-3.087	1.105	4.765	6.406	13.065	2.964	-0.659
QMJ	0.578	4.595	2.47	0.436	-15.232	-6.409	-4.755	0.458	5.782	8.367	18.382	1.485	0.135
ER2	0.348	3.411	2.00	0.354	-15.102	-4.260	-3.198	0.295	3.764	5.625	23.583	7.369	0.566
ER4	0.522	2.797	3.66	0.647	-10.400	-3.606	-2.697	0.458	3.787	4.867	22.383	9.444	1.088
ER8	0.571	2.748	4.07	0.720	-12.218	-3.278	-2.470	0.495	3.979	4.908	22.830	11.599	1.132
ER2(ALL)	0.676	3.034	4.37	0.772	-17.519	-3.588	-2.553	0.717	3.928	5.139	16.663	6.139	-0.452
ER4(ALL)	0.718	2.418	5.82	1.028	-11.228	-3.022	-2.181	0.755	3.516	4.391	15.155	4.290	0.132
ER8(ALL)	0.790	2.367	6.55	1.157	-7.977	-2.993	-1.895	0.863	3.528	4.410	16.323	4.781	0.573

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Panel B. Pearson Correlations between Monthly Factor Returns

	MKT	HML	SMB	RMW	CMA	UMD	I/A	ROE	BAB	QMJ	ER2	ER4	ER8
MKT-R_f	1.00												
HML	-0.26	1.00											
SMB	0.19	-0.19	1.00										
RMW	-0.36	0.34	-0.44	1.00									
CMA	-0.39	0.67	-0.08	0.18	1.00								
UMD	-0.18	-0.17	0.03	0.10	0.06	1.00							
I/A	-0.36	0.69	-0.18	0.27	0.91	0.03	1.00						
ROE	-0.30	0.08	-0.38	0.72	0.05	0.49	0.13	1.00					
BAB	-0.19	0.39	-0.05	0.41	0.35	0.25	0.38	0.36	1.00				
QMJ	-0.51	0.04	-0.56	0.73	0.10	0.26	0.14	0.66	0.18	1.00			
ER2	-0.12	0.76	-0.09	0.26	0.53	-0.42	0.56	-0.13	0.22	-0.04	1.00		
ER4	-0.13	0.74	-0.14	0.21	0.58	-0.29	0.62	-0.13	0.17	-0.04	0.88	1.00	
ER8	-0.13	0.68	-0.20	0.24	0.53	-0.30	0.62	-0.09	0.12	0.03	0.80	0.93	1.00
ER2(ALL)	-0.29								0.35	0.20			
ER4(ALL)	-0.28								0.30	0.15			
ER8(ALL)	-0.23								0.18	0.08			

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Table 5

Time-Series Regressions of Monthly Fundamental Factor Returns on Other Factors

The table reports results of time-series regressions of ER8 factor monthly returns (in percent) on those of other factors. FF3 is the Fama and French three-factor model, FF4 is that model with an added momentum factor, FF5 is the Fama and French five-factor model, and *q*CAPM is the *q*-investment CAPM. The factors are defined in Tables 2 and 4. ER8 is constructed from stocks excluding microcaps and ER8(ALL) from all stocks, including microcaps. The period runs from April 1982 to March 2014 or 384 months. t-statistics are in parentheses, and asterisks, ***, **, * indicate statistical significance at the 1%, 5%, and 10% level.

	ER8 MKT	ER8 FF3	ER8 FF4	ER8 FF5	ER8 *q*CAPM	ER8 BAB/QMJ	ER8(ALL) BAB/QMJ	ER8 All Factors
Intercept	0.627***	0.355***	0.452***	0.273***	0.315***	0.487**	0.671***	0.366***
	(4.21)	(3.26)	(3.37)	(2.44)	(2.30)	(2.39)	(3.85)	(3.01)
MKT-R_f	-0.079	0.036	0.009	0.068	0.048			0.051*
	(-1.24)	(0.88)	(0.31)	(1.62)	(1.16)			(1.93)
HML		0.639***	0.599***	0.524***				0.414***
		(10.17)	(14.23)	(6.29)				(6.66)
SMB		-0.074	-0.068	-0.084				-0.098
		(-1.30)	(-1.20)	(-1.32)				(-0.34)
UMD			-0.113					-0.048
			(-1.48)					(-1.15)
RMW				0.018				0.252*
				(0.23)				(1.77)
CMA				0.261**				-0.380*
				(2.37)				(-1.86)
ME					-0.144**			0.084
					(-2.15)			(0.27)
I/A					0.921***			0.890***
					(9.35)			(4.12)
ROE					-0.204*			-0.300**
					(-1.82)			(-2.04)
BAB						0.091	0.111**	-0.119***
						(1.33)	(2.44)	(-3.61)
QMJ						0.003	0.028	0.033
						(0.04)	(0.57)	(0.92)
Adj.R^2	0.014	0.472	0.503	0.488	0.434	0.010	0.096	0.613

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Table 6

Time-Series Regressions of Monthly Returns for Various Factors on the Fundamental Factor

The table reports results of time-series regressions of monthly returns (in percent) for factors in Table 5 on ER8 factor returns. The period is from April 1982 to March 2014 or 384 months. t-statistics are in parentheses, and asterisks, ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Factor	Intercept	ER8	Adj.R^2
MKT-R_f	0.825***	-0.214	0.014
	(3.21)	(-1.09)	
HML	-0.105	0.733***	0.467
	(-0.84)	(7.47)	
SMB	0.253	-0.214	0.037
	(1.52)	(-1.34)	
UMD	0.865***	-0.501*	0.089
	(4.23)	(-1.87)	
RMW	0.262	0.223	0.055
	(1.61)	(1.54)	
CMA	0.120	0.393***	0.283
	(1.29)	(4.31)	
ME	0.318*	-0.204	0.032
	(1.77)	(-1.19)	
I/A	0.145*	0.439***	0.384
	(1.83)	(6.16)	
ROE	0.605***	-0.082	0.005
	(4.56)	(-0.54)	
BAB	0.814***	0.167	0.013
	(3.31)	(1.02)	
QMJ	0.553**	0.044	-0.002
	(2.01)	(0.21)	

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Table 7

Fundamental Factor Returns During the Financial Crisis

The table reports monthly percentage returns from September 2008 through December 2009 for the CRSP Value-weighted Index (MKT), along with returns on the long side (ER-H) and short side (ER-L) of the ER8(ALL) fundamental factor and the factor return itself. Total monthly returns for two subperiods, the down-market and the up-market, are also reported.

Month	MKT	ER8-H	ER8-L	ER8
2008				
Sept	-9.09	-13.84	-14.07	0.23
Oct	-17.15	-24.25	-23.60	-0.65
Nov	-7.83	-17.59	-12.74	-4.85
Dec	1.74	3.58	5.74	-2.16
2009				
Jan	-8.12	-1.91	-2.52	0.61
Feb	-10.09	-11.28	-10.62	-0.66
DOWN				
Sept-Feb	-50.54	-65.29	-57.81	-7.48
Mar	8.97	13.45	10.93	2.52
Apr	10.20	30.42	14.10	16.32
May	5.21	11.61	6.95	4.66
June	0.44	4.95	3.29	1.65
July	7.73	12.48	8.18	4.30
Aug	3.34	8.28	1.73	6.55
Sept	4.09	9.74	7.55	2.19
Oct	-2.59	-5.77	-4.15	-1.62
Nov	5.56	2.87	4.37	-1.49
Dec	2.76	7.05	6.18	0.87
UP				
Mar-Dec	45.71	95.08	59.13	35.95

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Table 8

Tests of Two-factor Model Explaining ER Portfolio Returns

The left-hand side of the table reports coefficient estimates from time-series regression of monthly excess returns for ER_t decile portfolios on monthly excess returns for the CRSP value-weighted index and the ER8 fundamental factor (excluding microcaps). The right-hand side of the table reports on a similar regression, but with the ER8 factor return replaced by the excess return for the short side (ER8=L) of the ER8 factor. The period runs from April 1982 to March 2014 (384 months) with ER_t portfolios reformed each March 31. t-statistics are in parentheses, with asterisks indicating significance levels as in other tables.

Regression on left-hand side of the table: ER_t Portfolio Excess $Return_t = \alpha + \beta_1(MKT_t - R_{ft}) + \beta_2 ER8_t + e_t$

Regression of right-hand side of the table: ER_t Portfolio Excess $Return_t = \alpha + \beta_1(MKT_t - R_{ft}) + \beta_2(ER8\text{-}S_t - R_{ft}) + \varepsilon_t$

		Independent Variable=ER8 Factor				Independent Variable=ER8-L			
ER_t Decile	Mean Excess Return (%)	α	β_1	β_2	R^2	α	β_1	β_2	R^2
1	0.184	-0.003*	1.291***	-0.668***	0.760				
		(-1.82)	(31.72)	(-10.01)					
2	0.589	-0.000	1.200***	-0.430***	0.810				
		(-0.07)	(38.01)	(-8.31)					
3	0.718	0.001	1.154***	-0.334***	0.825				
		(0.73)	(40.50)	(-7.14)					
4	0.862	0.001	1.147***	-0.072*	0.847	0.002***	0.386***	0.615***	0.932
		(0.83)	(45.21)	(-1.74)		(2.61)	(10.00)	(22.03)	
5	0.973	0.002	1.127***	0.031	0.852	0.003***	0.436***	0.552***	0.926
		(1.45)	(46.52)	(0.77)		(4.05)	(11.11)	(19.48)	
6	0.993	0.002	1.071***	0.136***	0.839	0.004***	0.476***	0.469***	0.893
		(1.46)	(44.52)	(3.44)		(4.05)	(10.60)	(14.45)	
7	0.988	0.001	1.086***	0.269***	0.841	0.003***	0.499***	0.453***	0.875
		(0.64)	(44.86)	(6.77)		(3.59)	(10.20)	(12.80)	
8	1.159	0.001	1.163***	0.471***	0.822	0.005***	0.493***	0.507***	0.829
		(0.56)	(41.66)	(10.30)		(3.95)	(7.88)	(11.22)	
9	1.206	0.000	1.191***	0.622***	0.791	0.005***	0.509***	0.508***	0.764
		(0.09)	(37.28)	(11.87)		(3.42)	(6.56)	(9.06)	
10	1.486	0.000	1.369***	0.880***	0.699	0.007***	0.478***	0.658***	0.654
		(0.10)	(28.79)	(11.29)		(3.16)	(4.10)	(7.81)	

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INTERNET APPENDIX

Internet Appendix 1. Comparison of the Modeling with Existing Factor Models

A number of papers have attempted to model risk and expected return with accounting numbers, some of them with present-value pricing formulations. This appendix contrasts those papers with this one. The key difference is our recognition of accounting principles that connect accounting numbers to the risk to consumption. The modeling of accounting in some of the papers below is actually inconsistent with accounting principles, and other mistakes are made in handling accounting information.[29]

IA1.1 Fama and French Reverse Engineering

In developing their five-factor model, Fama and French (2015) refer to a valuation model with expected earnings and book value. With a constant required return, *r* (and the expectation operator implied for $\tau > t$):

$$\frac{P_t}{B_t} = \frac{\sum_{\tau=t+1}^{\infty} (Earnings_\tau - (B_\tau - B_{\tau-1})/(1+r)^\tau}{B_t} \quad . \tag{A1}$$

They refer to the term, $\frac{\sum_{\tau=t+1}^{\infty} Earnings_\tau}{B_t}$ (the infinite horizon expected book rate of return), as "profitability" with a measure of the current book rate of return their choice of the indicator (and the basis for their profitability factor). The change in book value, $\frac{B_\tau - B_{\tau-1}}{B_t}$, is given the label, "growth in investment." Reverse-engineering the expression, and carrying out some comparative statics, they conclude that (i) *r* is increasing in B/P holding the expected book rate of return and expected investment growth constant, (ii) *r* is increasing in the expected book rate of return holding B/P and expected investment growth constant, and (iii) *r* is decreasing in expected investment growth holding B/P and expected book rate of return constant. B/P, return on book value, and investment then become characteristics from which factors are formed.

However, the comparative statics are inconsistent with how accounting works. One cannot vary B/P holding expected book rate of return constant. Book value denominates expected future book rate of return—a lower book value (and B/P) yields a higher expected return on book value, by construction of the accounting. That accounting is what we focus on in this paper: Accounting that defers earnings to the future reduces current book value (by expensing investment to the income statement rather than booking them to the balance sheet, for example) and increases expected future earnings which, on the lower book value, results in a higher expected future book rate of return. In short, expected book rate of return and B/P vary inversely, *ceteris paribus* (as empirical research confirms). Further, while expected future profitability on current book value could indicate expected returns, current book rate of return cannot be an indicator of the

[29] Portions of the critique are in Penman (2016), Penman, Reggiani, Richardson, and Tuna (2018), and Penman and Yehuda (2019, appendix).

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future rate of return because some of the investment that yields the expected future earnings is expensed, yielding lower current earnings and thus lower book rate of return but higher expected return in the future. And that expensing of investment yields a lower denominator for the expected future book rate of return, and thus higher expected profitability against lower current book rate of return. Penman and Zhang (2021) provide a further critique of the profitability factor.

$\frac{B_\tau - B_{\tau-1}}{B_t}$ is held constant in the comparative statics with the label, "investment growth." This is growth in shareholders' equity, not growth of investment in the business which can also be financed by borrowings. And $B_\tau - B_{\tau\text{-}1} = Earnings_\tau - d_\tau$ by CSR, and payout is not relevant to price under M&M assumptions: Growth in equity investment from retention, $Earnings_\tau - d_\tau$, does not add value, nor does it affect price and B/P, unless the retained earnings are invested in value-adding investments in the business. Further, given $Earnings_\tau$, dividends or stock repurchases financed by borrowings changes $B_\tau - B_{\tau\text{-}1}$ but does not affect that investment. By referring to investment growth, the analysis nominally introduces growth, but there is a mislabeling that voids the interpretation.

The Fama and French comparative statics appear to support the view, common in asset pricing, that expected returns are negatively related to growth. Definitions of "growth" may vary, but our analysis suggests that the expected return is positively related to expected earnings growth. With future book rate of return defined over an infinite forecast horizon as $\frac{\sum_{\tau=t+1}^{\infty} Earnings_\tau}{B_\tau}$ and current book return taken as a sufficient statistic for that future book rate of return, the Fama and French formulation does not admit the effect of accounting principles on intertemporal earnings recognition nor period-to-period earnings growth. In contrast, our analysis focuses on the recognition of earnings in the short-term versus the long-term under accounting that reduces current profitability and increases expected earnings growth. Equation (A1) is silent with respect to accounting principles—random numbers for earnings will satisfy the equation. One cannot get traction in tying accounting numbers to risk and return without further specification of accounting principles.

IA1.2 The Vuolteenaho (2002) Autoregressive Accounting Model

Vuoteenaho (2002) applies the clean surplus relation to extract "cash-flow news" and "discount-rate news" from the evolution of accounting numbers, specifically B/P and ROE. Lyle and Wang (2015) embrace the Vuolteenaho modeling to estimate expected returns.

The critique of the Vuolteenaho modeling is in internet appendix to Penman and Yehuda (2019), at www8.gsb.columbia.edu/researcharchive/articles/25639 to which the reader is referred. Also see Penman (2016). There are three main points. First, ROE and B/P are affected by net dividends, so the evolution of these numbers from which discount-rate news and expected returns are extracted is partially driven by dividends. Thus there is a violation of the M&M dividend irrelevance proposition. Second, the model assumes an autoregressive process to describe the evolution of ROE and B/P. The evolution of accounting numbers is determined by the accounting principles governing those numbers and an autoregressive process is inconsistent

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with the accounting principles governing earnings and book value, namely the principles that are invoked in this paper. Thus the modeling does not imbed accounting principles that tie earnings to consumption and the risk to consumption. Surprisingly, the modeling sees earnings realizations as indicating higher risk, the opposite to the modeling here. Third, empirically the modeling fails consistency conditions required of estimates that indicate expected returns.

Campbell, Polk, and Vuolteenaho (2010) also apply the autoregressive modeling to explain stocks' systematic risk as driven by risk to these fundamentals. In one test, they assume that ROE conveys cash-flow news while the price-earnings ratio (P/E) conveys expected return news. That, is inconsistent with accounting principles under which ROE conveys risk and expected return (here and in Penman and Zhang 2019) and P/E imbeds both earnings growth and the risk to that growth. In their vector autoregression (VAR) tests, the paper is careful to say that the results are conditional upon their specification of the VAR. The inconsistency of VAR with accounting principles is the point at issue.

IA1.3 The Investment (q-Factor) CAPM

The *q*-investment CAPM of Hou, Xue, and Zhang (2015) and its q^5 extension in Hou, Mo, Xue, and Zhang (2020) introduces investment and book rate of return (ROE) to explain expected returns. Importantly, the models are based on theory, on economic modeling of production rather than data mining. They reach the conclusion that high expected profitability for given investment implies a higher expected return, more so with low investment. However, in applying the models, the papers employ accounting numbers for investment and ROE that differ from the constructs in mind (as the authors recognize).

Under accounting principles, investment on the balance sheet, the measure adopted in the papers, does not represent investment—only the part of investment with a relatively certain payoff (and a small amount of total investment for the modern corporation). The remainder, particularly risky investment, is expensed against earnings. Investment booked to the balance sheet is associated with lower expected returns (Table 1 Panel A) but that which is expensed to the income statement is associated with higher expected returns, both in theory in Penman and Zhang (2020) and empirically in Penman and Zhang (2021). The latter contrasts to the negative correlation in the investment models. That, it turn, affects ROE, inducing a negative relation between ROE and returns so affected, not the positive relation predicted by the investment models.

IA1.4 The Quality Factor

A "quality factor" (QMJ), introduced by Asness, Frazzini, and Pedersen (2019) is built from fundamental accounting numbers. The idea behind it appears to be to define "good quality companies," with accounting measures such as profitability, asset turnover, accruals, earnings quality, (low) earnings variability, and (low) leverage going into the factor construction. These are features that one might well associate intuitively with risk, but no formal connection is made. Oddly, while these are features that one might well associate with less risk, "quality" appears to yield higher returns. There is no formal modeling; the justification is that it seems to work in the data. Frankly, we do not understand why (fundamentally).

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Internet Appendix 2. Results with Value-weighted Portfolios

In the paper, ER portfolios are equally-weighted. Value-weighting is inappropriate because expected earnings growth at risk is less prevalent in large firms, as reported in Penman, Reggiani, Richardson, and Tuna (2018) and Penman and Reggiani (2018, 2020). Thus value weighting weights away from the exposure which the fundamental factor is designed to capture to firms that are less sensitive to it. Consequently, value weighted ER portfolios do not capture the sensitivity of returns to the underlying risk. For example, if a large firm were exposed to this risk (as some are), sensitivity of its returns to a value-weighted ER portfolio return would not pick up the exposure. The risk in a small firm would similarly not be calibrated.

To demonstrate the point, the tables here repeat Panel A of Table 5, Table 6, and Table 7 but with value weighting in the ER long and short portfolios. The pertinent mean returns, t-statistics, Sharpe ratios, and return kurtosis for the ER factors are lower than those in Table 5 due to the emphasis on large firms. Correspondingly, with the weight shifted away from growth at risk, the tests of fundamental factor model against other factor models in Tables 6 and 7 are not as favorable to the factor. It is clear that the results with equal weighting are due to relatively more weight given to small firms where the feature underlying the portfolio formation, growth at risk, is stronger.

Most of the comparison factors use value weighting in their construction. (The BAB factor weights stocks on estimated betas, not market value.) So the question arises as to whether these factors also are weighting away from growth at risk. For example, B/P is correlated with ER in Tables 1 and 5, indicating it picks up the exposure. But value weighting in the HML factor weights away from the risk. (The size factor in the models may ameliorate.) The issue is a little more complex for, in certain periods, the risk is associated with large firms and these are periods where it has been documented that the "size effect in returns" reverses, as does the return to the SML size factor. See Penman and Reggiani (2020) for the documentation.

Value-weighted portfolios are sometimes justified because they are viewed as more tradable. The paper accommodates the point by reporting results elimination microcaps (below the 20^{Th} percentile of NYSE stocks by size) where liquidity may be an issue. Value weighting is also often justified to avoid monthly rebalancing, but this is not a concern for annual returns on portfolios reformed annually based on updated annual accounting information.

Value-weighing is only one issue that arises in comparisons with alternative factors, for some of those factors also depend on portfolio construction: The subset of assets used, breakpoints for portfolio formation, the order of conditional sorting, independent versus nested sorts, and frequency of rebalancing, to name a few. These issues have been much discussed in the literature, for example whether the Fama and French sorts separate size and B/P appropriately. So, any comparison is subject to wider issues in the comparison portfolios.

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Table A1: Table 4, Panel A in the Paper but with ER Factors Value Weighted

Summary Statistics for Monthly Percentage Factor Returns

Factor	Mean	Std Dev	t-value	Annual Sharpe Ratio	Min	p5	p10	p50	p90	p95	Max	Kurtosis	Skew
MKT-R_f	0.703	4.506	3.06	0.541	-23.240	-7.260	-4.780	1.170	6.080	7.190	12.470	2.578	-0.789
HML	0.313	2.944	2.08	0.368	-11.250	-3.910	-2.890	0.170	3.780	5.440	12.910	2.482	0.099
SMB	0.131	2.946	0.87	0.154	-15.280	-4.070	-3.290	0.000	3.430	4.640	18.730	5.709	0.478
RMW	0.389	2.556	2.98	0.527	-19.110	-3.050	-1.850	0.410	2.940	3.880	13.520	12.786	-0.513
CMA	0.345	2.025	3.34	0.590	-6.880	-2.540	-1.800	0.195	2.900	3.310	9.550	2.183	0.422
UMD	0.579	4.562	2.49	0.440	-34.580	-6.800	-4.710	0.685	4.880	6.540	18.380	12.447	-1.651
I/A	0.396	1.941	4.00	0.706	-7.160	-2.490	-1.880	0.365	2.690	3.350	9.250	1.859	0.180
ROE	0.558	2.621	4.17	0.737	-13.850	-3.670	-2.050	0.655	3.260	4.020	10.380	5.100	-0.773
BAB	0.909	3.708	4.80	0.849	-15.132	-6.022	-3.087	1.105	4.765	6.406	13.065	2.964	-0.659
QMJ	0.578	4.595	2.47	0.436	-15.232	-6.409	-4.755	0.458	5.782	8.367	18.382	1.485	0.135
ER2	0.157	3.415	0.90	0.159	-10.905	-4.907	-3.436	-0.017	3.899	6.352	19.547	4.177	0.762
ER4	0.317	2.955	2.10	0.371	-6.979	-4.170	-2.974	0.109	3.863	4.957	17.723	3.410	0.897
ER8	0.373	2.768	2.64	0.467	-7.680	-4.040	-2.619	0.171	3.699	5.309	19.122	5.817	1.067
ER2(ALL)	0.225	3.378	1.30	0.230	-10.219	-4.805	-3.640	0.256	3.853	5.322	21.870	4.960	0.877
ER4(ALL)	0.320	2.973	2.10	0.372	-8.093	-4.091	-3.284	0.372	3.739	5.130	17.831	3.184	0.708
ER8(ALL)	0.465	2.911	3.13	0.554	-9.205	-3.765	-2.806	0.336	3.873	4.649	21.153	7.206	1.080

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Table A2: Table 5 in the Paper but with ER Factor Value Weighted

Time-Series Regressions of Monthly Fundamental Factor Returns on Other Factors

	ER8	ER8	ER8	ER8	ER8	ER8	ER8 (ALL)	ER8
Intercept	0.432***	0.166	0.288**	0.078	0.188	0.309*	0.464**	0.174
	(2.89)	(1.41)	(2.30)	(0.66)	(1.54)	(1.69)	(2.59)	(1.34)
MKT-R_f	-0.084	0.026	-0.009	0.061	0.026			0.076***
	(-1.46)	(0.73)	(-0.34)	(1.59)	(0.73)			(2.84)
HML		0.623***	0.573***	0.485***				0.429***
		(12.50)	(11.69)	(6.44)				(5.95)
SMB		-0.049	-0.040	-0.070				0.124
		(-0.95)	(-0.87)	(-1.13)				(0.53)
UMD			-0.143**					-0.075*
			(-2.54)					(-1.78)
RMW				-0.008				0.174
				(-0.12)				(1.63)
CMA				0.328***				-0.119
				(2.63)				(-0.72)
ME					-0.147**			-0.089
					(-2.31)			(-0.38)
I/A					0.887***			0.647***
					(9.99)			(3.71)
ROE					-0.278***			-0.334***
					(-2.95)			(-2.73)
BAB						0.059	0.044	-0.125***
						(0.77)	(0.57)	(-3.39)
QMJ						0.019	-0.056	0.128***
						(0.29)	(-1.02)	(3.03)
Adj.R2	0.016	0.437	0.487	0.463	0.426	0.003	0.006	0.608

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Table A3: Table 6 in the Paper with ER Factor Value Weighted

Time-Series Regressions of Monthly Returns for Various Factors on the Fundamental Factor

Factor	Intercept	ER8	Adj.R^2
MKT-R_f	0.786***	-0.223	0.016
	(3.29)	(-1.36)	
HML	0.051	0.704***	0.436
	(0.38)	(8.97)	
SMB	0.199	-0.182	0.027
	(1.27)	(-1.32)	
UMD	0.787***	-0.557**	0.112
	(3.99)	(-2.45)	
RMW	0.318**	0.190	0.040
	(2.16)	(1.34)	
CMA	0.195**	0.402***	0.299
	(2.20)	(5.68)	
ME	0.271	-0.187	0.027
	(1.65)	(-1.27)	
I/A	0.240***	0.419***	0.355
	(2.85)	(8.05)	
ROE	0.610***	-0.139	0.019
	(4.99)	(-0.97)	
BAB	0.867***	0.112	0.004
	(3.72)	(0.71)	
QMJ	0.550**	0.075	-0.001
	(2.13)	(0.37)	

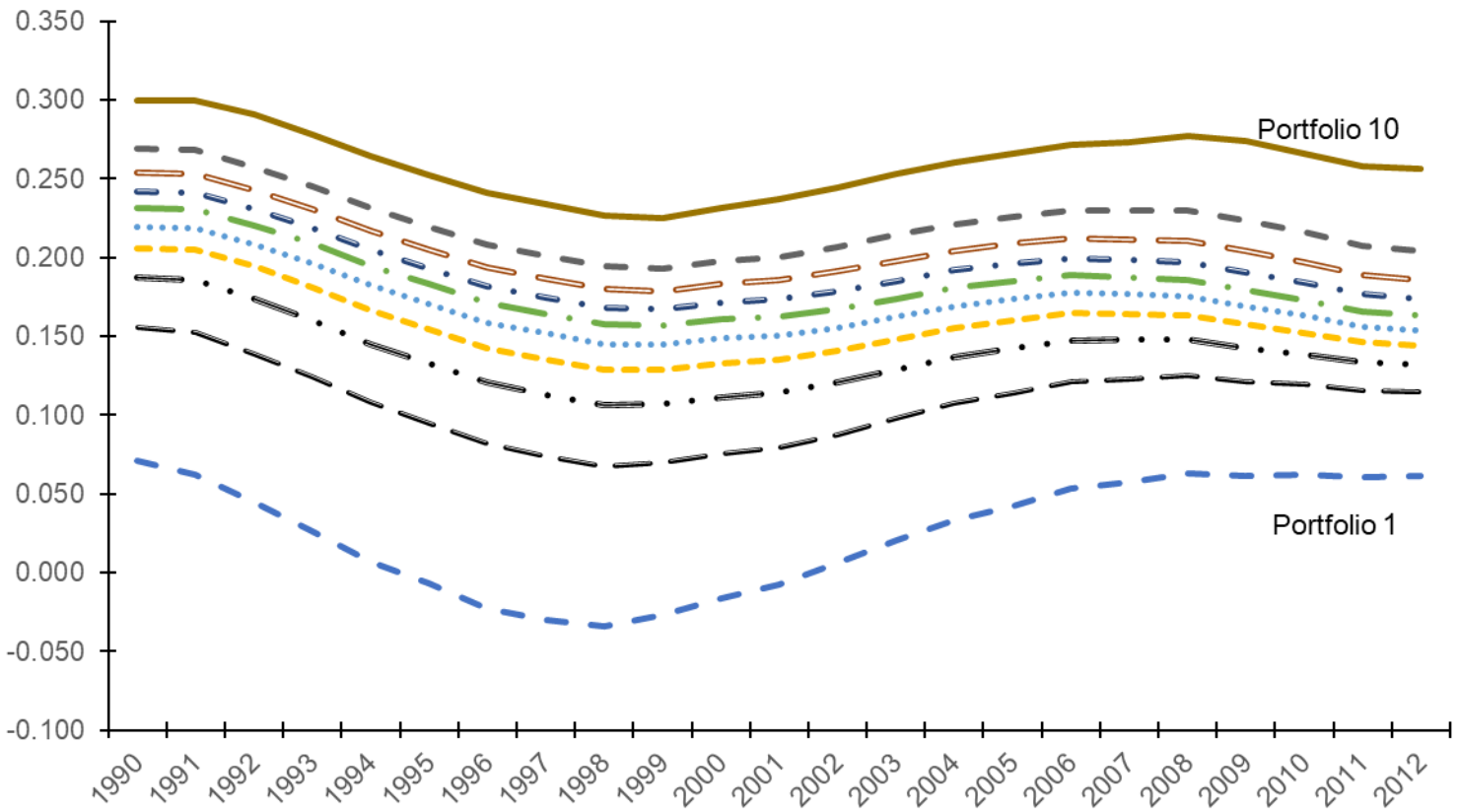
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Internet Appendix 3. Time-series Variation in Expected Returns, ER_t

The cross-sectional variation in portfolio ER_t is evident in the paper. The time-series standard deviation of the yearly portfolio ER_t estimates is close to 0.04 for most portfolios, except for portfolio 1 where it is 0.056. We do not have a benchmark for how ER_t should vary over time (with changes in interest rates and risk premia), and some of the variation is presumably due to measurement error in the ER_{it}. That said, this variation is not large enough to suggest that the estimates are wildly fluctuating. Indeed, they are slowly moving.

Figure A1 plots the variation of the 10-year moving average of ER_t over time, for 1990-2012. The averaging washes out mean-zero estimation error if the "true" expected return is changing slowly. It is clear that the differences in ER_t between portfolios is maintained over time. Further, the estimates co-vary over time, indicating they share common varying systematic risk. For all portfolios, the estimates are lower in the mid-to-late 1990s, a period of quite high ex post returns. The seemingly low ER_t estimates for portfolio 1 in Table 1 in the paper are due mainly to this period when they were negative. (Some commentators conjecture that the mid-to-late 1990s was a period when risk premiums declined.) But, relative to the variation across portfolios, the year-to-year estimates are quite stable for all portfolios, as are estimates averaged over also consecutive 2-year and 3-year periods.

Figure 1. Ten-year Rolling Averages of Estimated Expected Returns, ER_t, for Ten ER_t Portfolios, 1990-2012



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Internet Appendix 4. The Two-Factor Model for Fama and French Size and book-to-Market Portfolios.

This appendix reports on the same regressions as those in Table 8 in the paper, but for the 25 Fama and French portfolios that interact size (ME) and book-to-market (BE/ME) in the Kenneth French Data Library. Results are similar: The returns for these portfolios are explained by the two-factor model.

In Table A4 there is little ordering of the mean monthly excess returns over size portfolios 1 – 5, consistent with the observations that there has not been much of a "size effect" since 1980 when our sample period begins. Within size groups, BE/ME ranks mean returns. However, with the exception of high BE/ME for size portfolio 1 (large firms), the intercepts for all assets in the regressions on the left-hand side of the table are not significantly different from zero. Like Table 8, the returns on most of the BE/ME portfolios can be explained by exposure to the market and the ER fundamental factor. Betas on the market portfolio are similar over BE/ME portfolios for a given size portfolio and close to 1.0, though somewhat higher for some low BE/ME portfolios (portfolio 1). However, the mean coefficient on the ER factor is increasing over BE/ME. Thus, increasing returns to increasing BE/ME are explained by the same exposure to the market factor but an increasing exposure to the ER factor.

Higher exposure to ER means less insurance against a loss to consumption. The regressions on the right-hand side of Table A4 confirm. As in Table 8, the market excess return and the return on ER-L are the explanatory variables, with the ER-L portfolio the insurance. Unlike the corresponding regressions in Table 8, the (conditional) betas on the market return are increasing with BE/ME: Given ER-L, higher book-to-price indicates higher (conditional) exposure to market risk. However, the sensitivity coefficient to the hedge, ER-L, declines as BE/ME increases, and the mean intercepts increase significantly: The higher mean returns to BE/ME are explained by increasing sensitivity to the market with less protection from the hedge. As in Table 8, $\beta_1 + \beta_2$, roughly sum to 1.0 for all portfolios. The increasing intercepts capture the higher risk conveyed by BE/ME. The comparison of these intercepts with the zero intercepts for the left-hand side regressions indicate that BE/ME captures exposure to the (risky) long side of ER. (HML and ER are positively correlated 0.68 in T4B.) With respect to size, these patterns differ for the smallest firms (ME = 5) where (zero) intercepts, the market betas, and ER-L betas differ little over BE/ME but returns are still explained by the market factor and ER-L. Results are similar with equally weighted Fama and French portfolio returns and with ER8-ALL (based on all stocks, including microcaps).

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Table A4. Two-factor Model Explaining Fama and French Size and Book-to-Market Portfolio Returns

This table repeats the time-series regressions in Table 8 in the paper, but now for Fama and French Size and Book-to-Market factor portfolios. Size (ME) portfolio 1 are the largest firms, portfolio 5 the smallest. BE/ME portfolio 1 are lowest book-to-price firms, portfolio 5 the highest. As in Table 8, the left-hand side of the table reports coefficient estimates from time-series regression of monthly excess returns for Fama and French 25 size (ME) and book-to-price (BE/ME) value-weighed portfolios on excess monthly returns for the CRSP value-weighted index and those for the ER8 fundamental factor. The right-hand side of the table reports on a similar regression, but with the ER8 factor return replaced by the return for the short side (ER-L) of the ER8 factor. The period runs from April 1982 to March 2014 (384 months) with ER_t portfolios reformed each March 31. t-statistics are in parentheses under coefficient estimates. The portfolio returns are from the Kenneth French website library.

Regression on left-hand side of the table: FF25 ME-BE/ME Portfolio Excess $Return_t = \alpha + \beta_1(MKT_t - R_{ft}) + \beta_2 ER8_t + e_t$
Regression of right-hand side of the table: FF25 ME-BE/ME Excess $Return_t = \alpha + \beta_1(MKT_t - R_{ft}) + \beta_2(ER8\text{-}L_t - R_{ft}) + e_t$

			Independent Variable=ER8 Factor				Independent Variable=ER8-L			
ME Group	BE/ME Group	Mean Excess Return (%)	α	β_1	β_2	R^2	α	β_1	β_2	R^2
1	1	0.602	-0.003	1.269	-0.529	0.525	-0.003	-0.405	1.376	0.692
			(-1.10)	(19.14)	(-4.87)		(-1.39)	(-3.32)	(15.61)	
1	2	1.243	0.003	1.102	-0.266	0.573	0.004	-0.251	1.102	0.756
			(1.15)	(21.75)	(-3.20)		(2.13)	(-2.87)	(17.41)	
1	3	1.370	0.003	1.002	-0.035	0.605	0.005	-0.091	0.879	0.771
			(1.73)	(23.86)	(-0.51)		(3.63)	(-1.25)	(16.65)	
1	4	1.516	0.005	0.886	0.051	0.575	0.007	-0.033	0.733	0.717
			(2.84)	(22.59)	(0.79)		(4.99)	(-0.45)	(13.88)	
1	5	1.774	0.007	0.916	0.161	0.509	0.010	-0.019	0.740	0.625
			(3.23)	(19.87)	(2.13)		(5.30)	(-0.21)	(11.11)	
2	1	0.816	-0.002	1.365	-0.538	0.747	-0.002	-0.108	1.215	0.899
			(-0.92)	(31.39)	(-7.55)		(-1.84)	(-1.72)	(26.81)	
2	2	1.234	0.001	1.178	-0.055	0.759	0.003	0.066	0.895	0.915
			(0.55)	(34.23)	(-0.97)		(2.91)	(1.42)	(26.53)	
2	3	1.407	0.002	1.049	0.212	0.754	0.005	0.258	0.621	0.840
			(1.41)	(34.21)	(4.22)		(4.30)	(4.55)	(15.19)	
2	4	1.374	0.001	1.003	0.313	0.747	0.005	0.368	0.489	0.785
			(1.02)	(33.50)	(6.38)		(3.64)	(5.83)	(10.73)	
2	5	1.366	-0.001	1.171	0.533	0.719	0.004	0.419	0.569	0.726
			(-0.63)	(30.92)	(8.58)		(2.13)	(4.90)	(9.21)	

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3	1	0.992	-0.000	1.325	-0.455	0.784	-0.001	0.142	0.978	0.893
			(-0.16)	(35.13)	(-7.36)		(-0.65)	(2.34)	(22.29)	
3	2	1.312	0.001	1.145	0.080	0.813	0.003	0.375	0.613	0.896
			(0.86)	(40.48)	(1.73)		(3.24)	(7.77)	(17.56)	
3	3	1.254	0.000	1.040	0.243	0.807	0.003	0.495	0.422	0.840
			(0.29)	(39.95)	(5.70)		(2.68)	(9.11)	(10.75)	
3	4	1.402	0.001	0.993	0.400	0.780	0.005	0.561	0.321	0.763
			(1.00)	(36.47)	(8.96)		(3.55)	(8.68)	(6.88)	
3	5	1.575	0.001	1.110	0.572	0.751	0.006	0.619	0.357	0.707
			(0.77)	(33.34)	(10.48)		(3.44)	(7.50)	(5.98)	
4	1	1.178	0.002	1.228	-0.351	0.858	0.001	0.416	0.674	0.920
			(1.34)	(45.81)	(-7.99)		(1.19)	(9.06)	(20.32)	
4	2	1.231	0.000	1.093	0.188	0.839	0.002	0.654	0.340	0.859
			(0.05)	(44.51)	(4.68)		(1.98)	(12.47)	(8.96)	
4	3	1.174	-0.002	1.076	0.411	0.821	0.002	0.737	0.246	0.793
			(-1.38)	(41.64)	(9.70)		(1.20)	(11.58)	(5.35)	
4	4	1.343	0.000	1.002	0.427	0.819	0.003	0.830	0.111	0.768
			(0.41)	(41.14)	(10.69)		(2.72)	(13.17)	(2.43)	
4	5	1.315	-0.003	1.170	0.722	0.796	0.002	0.993	0.096	0.691
			(-1.85)	(37.38)	(14.06)		(1.19)	(11.29)	(1.52)	
5	1	1.075	0.000	1.116	-0.118	0.916	-0.000	0.947	0.143	0.917
			(0.11)	(63.17)	(-4.08)		(-0.38)	(23.67)	(4.96)	
5	2	1.206	0.000	1.012	0.173	0.879	0.002	0.939	0.048	0.870
			(0.52)	(52.66)	(5.49)		(1.84)	(20.63)	(1.44)	
5	3	1.233	0.000	0.940	0.366	0.815	0.002	1.040	-0.104	0.773
			(0.13)	(40.67)	(9.65)		(1.88)	(17.79)	(-2.46)	
5	4	1.073	-0.003	0.962	0.551	0.787	0.000	1.094	-0.142	0.701
			(-2.19)	(36.61)	(12.81)		(0.32)	(15.39)	(-2.76)	
5	5	1.345	-0.001	1.053	0.707	0.743	0.003	1.220	-0.179	0.635
			(-0.96)	(31.93)	(13.07)		(1.42)	(13.58)	(-2.76)	

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